

The Course

Synthetic Aperture Radar (SAR)

Part 2: Azimuth Resolution

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

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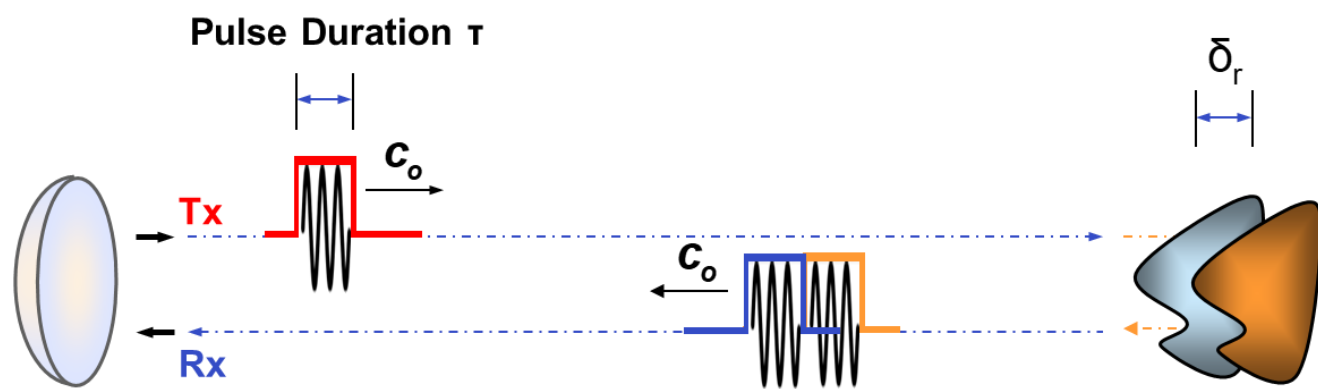
ESAMAAP

EEBI  **MASS**

Azimuth = Platform Propagation Direction

Range = Radar Illumination Direction





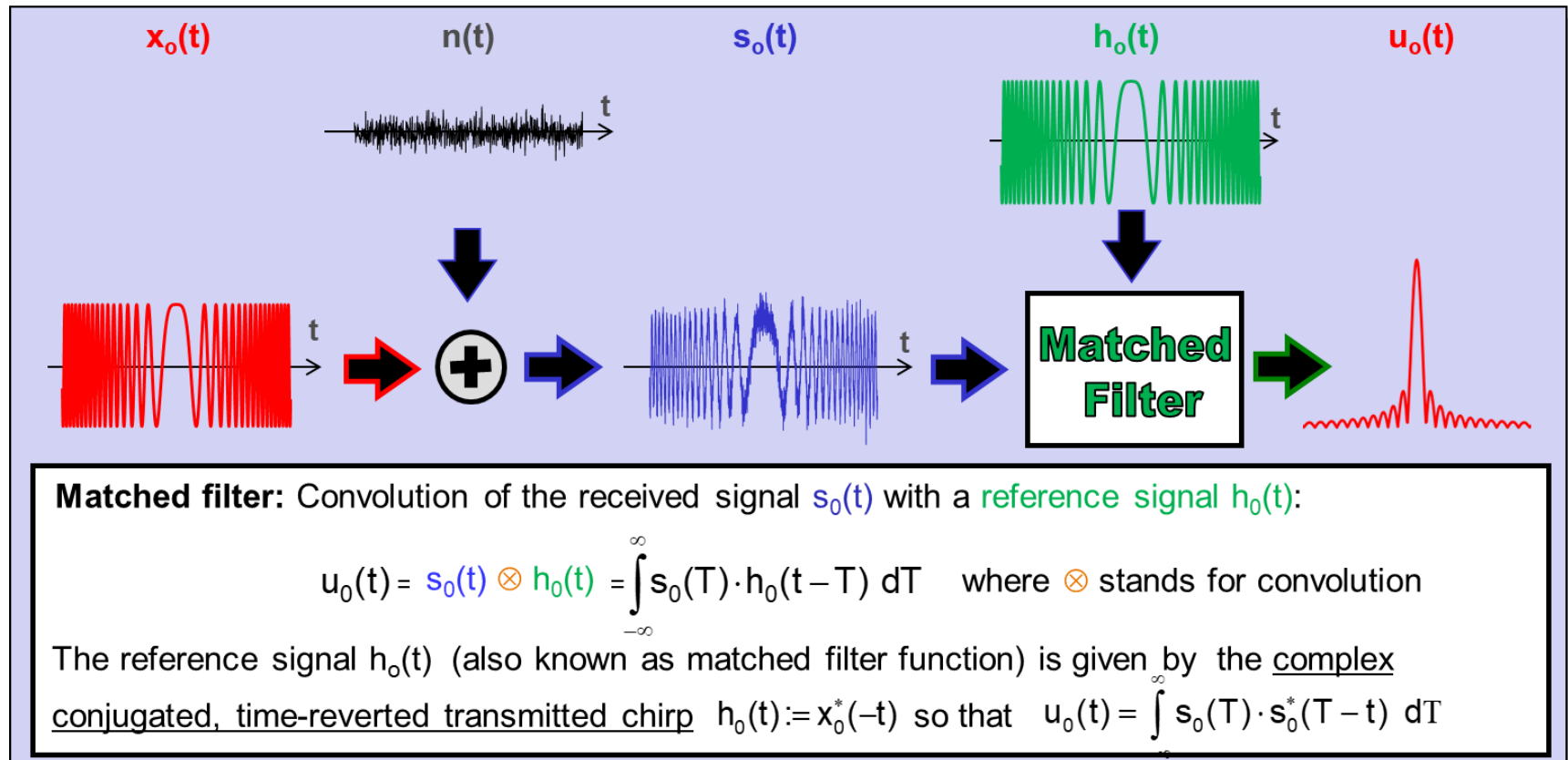
Range Resolution

A pulse bandwidth of $W=100\text{MHz}$ leads to a range resolution of $\delta_r=1.5\text{m}$

This corresponds to a pulse duration (after compression) of $\tau=10\text{ns}$.

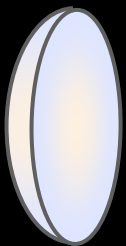
The range resolution is independent on the distance between the sensor and the scatterer; it only depends on the pulse duration τ that is inverse proportional to the pulse bandwidth W : $\delta_r = \frac{c\tau}{2} = \frac{c}{2W}$

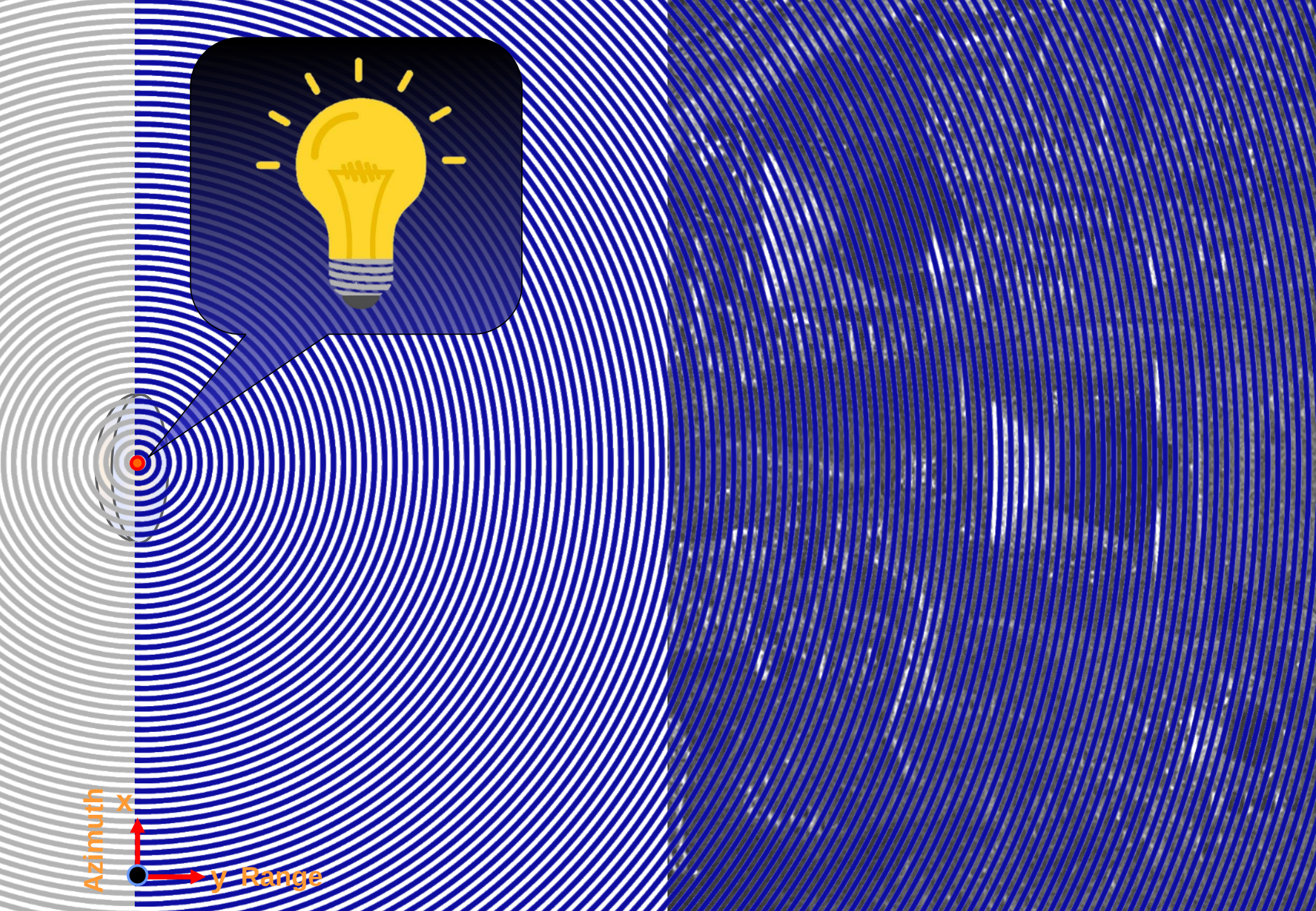
Pulse Compression (Matched Filtering)

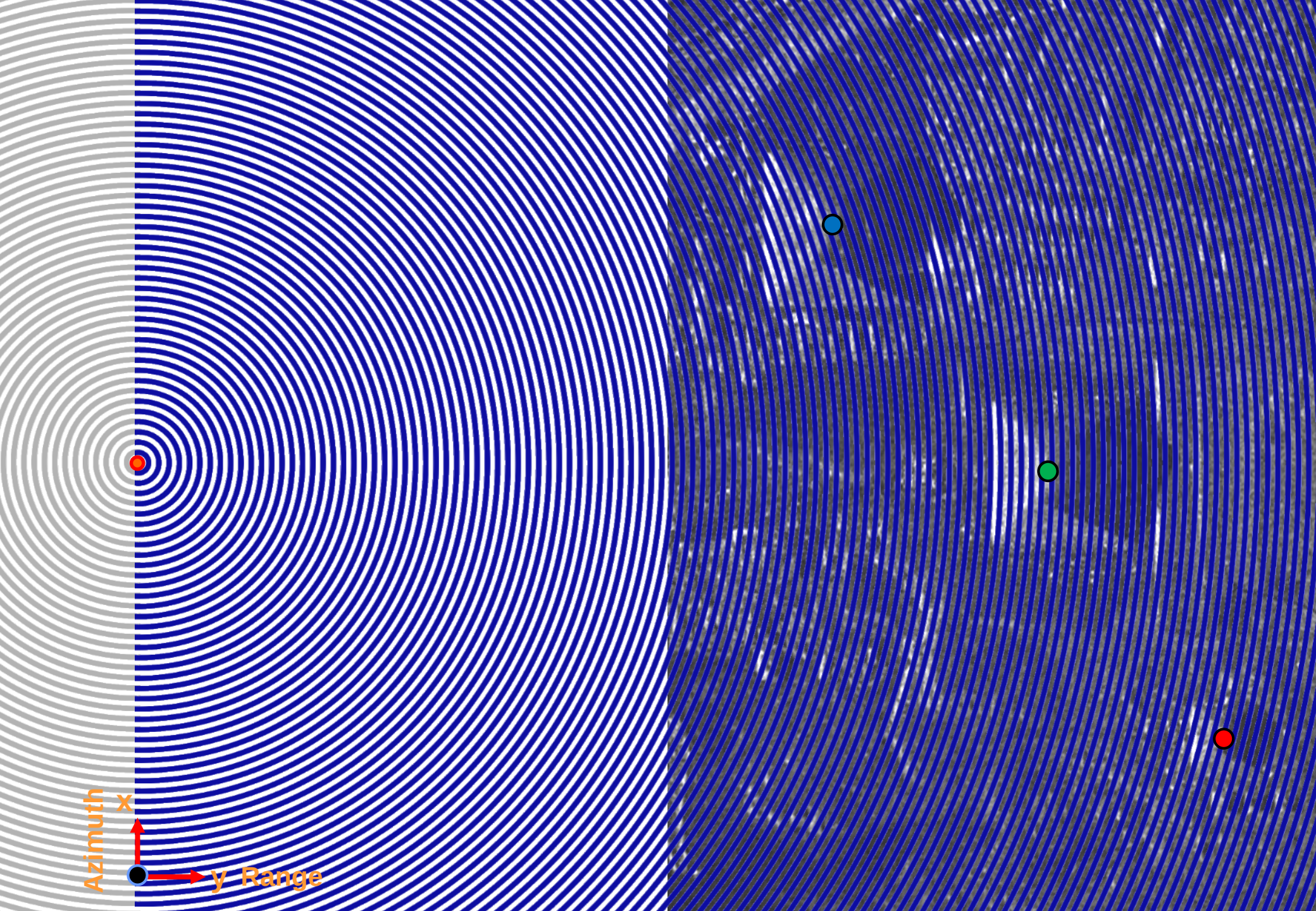


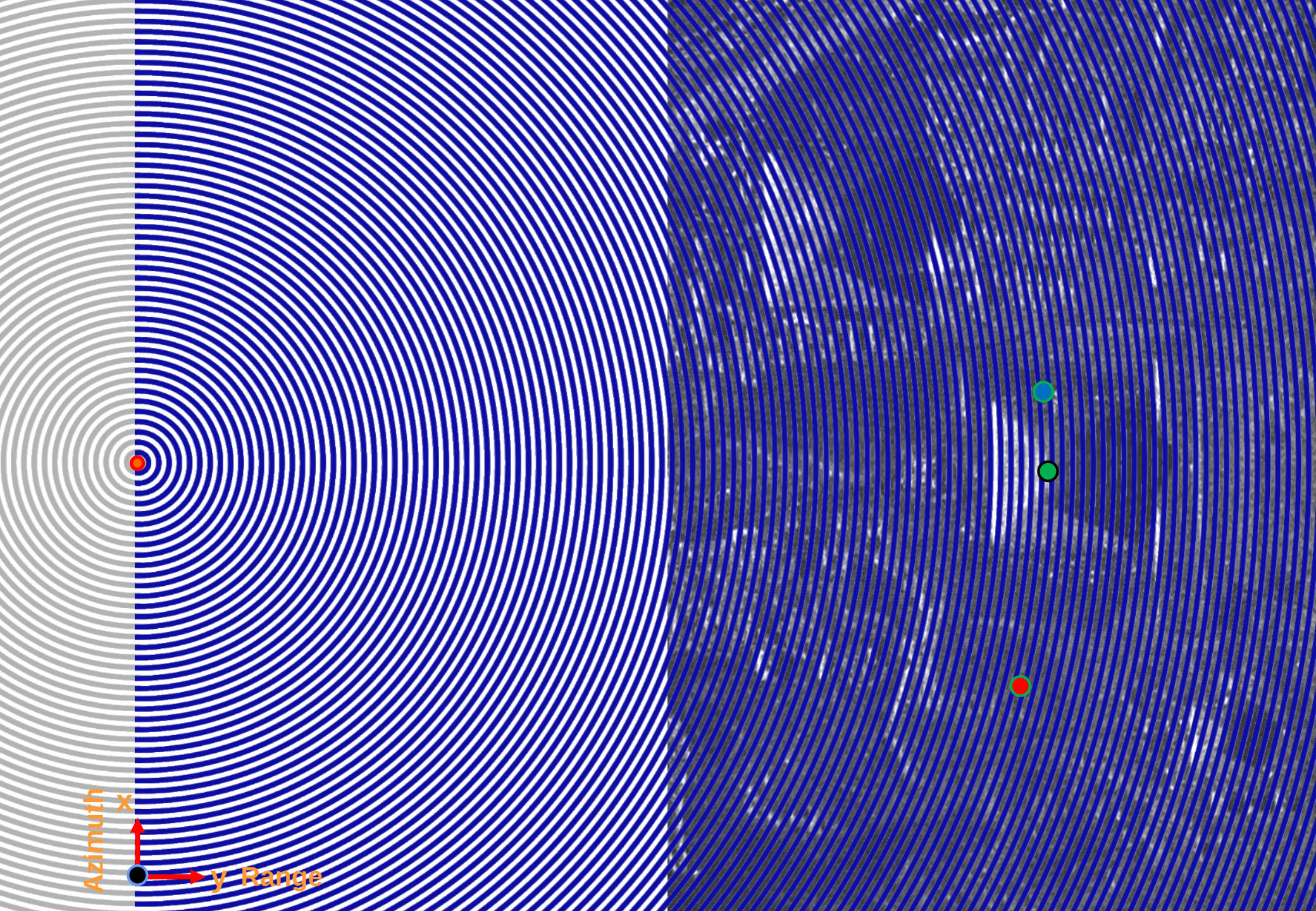


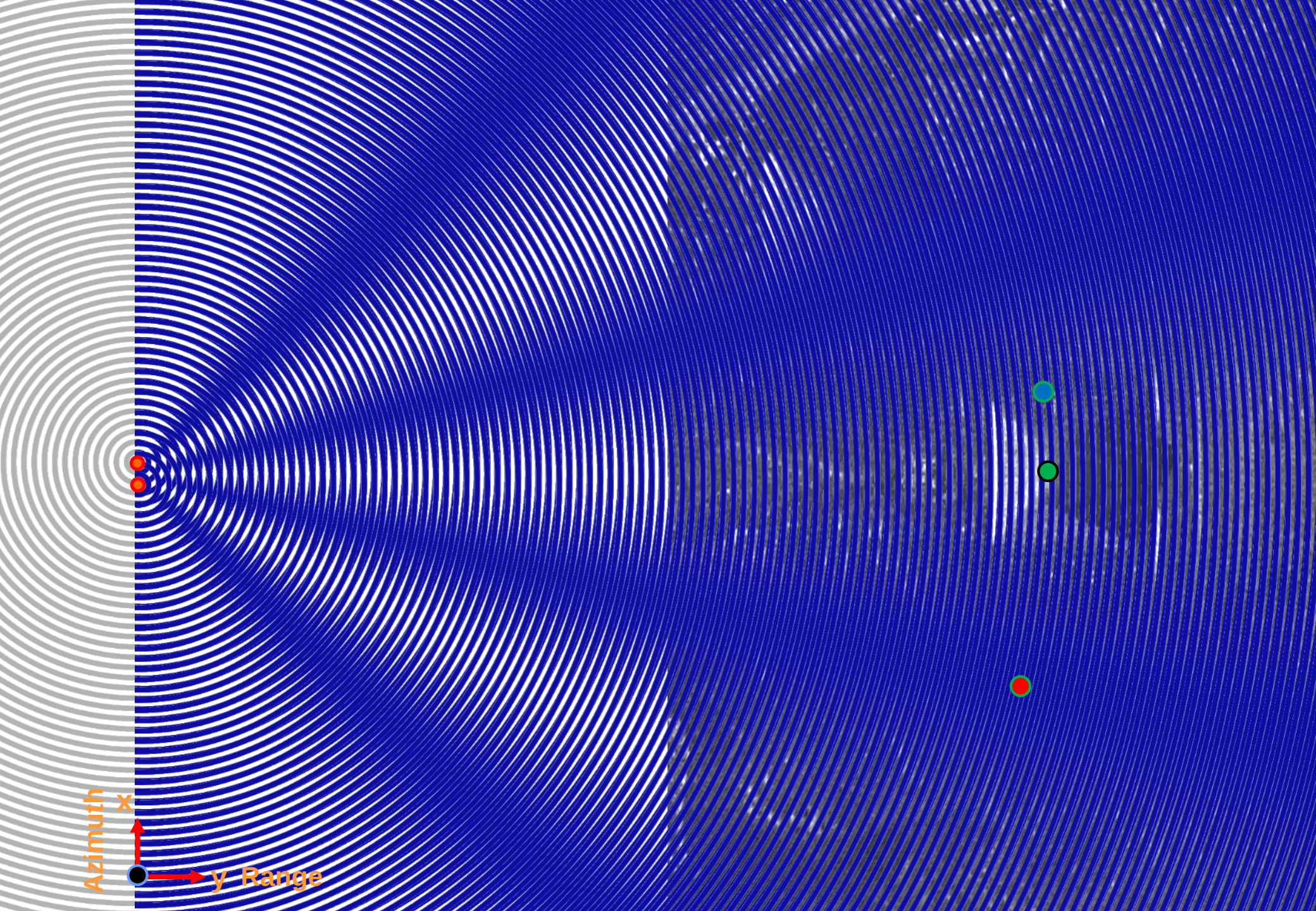
Azimuth Resolution











Antenna
Range

Azimuth
x
y Range



Angular resolution: $\theta_a \approx \frac{\lambda}{d_a} = \frac{\text{Wavelength}}{\text{(Real) Antenna Aperture}}$

(Diffraction)

Antenna
Real

d_a

θ_a

Azimuth

x

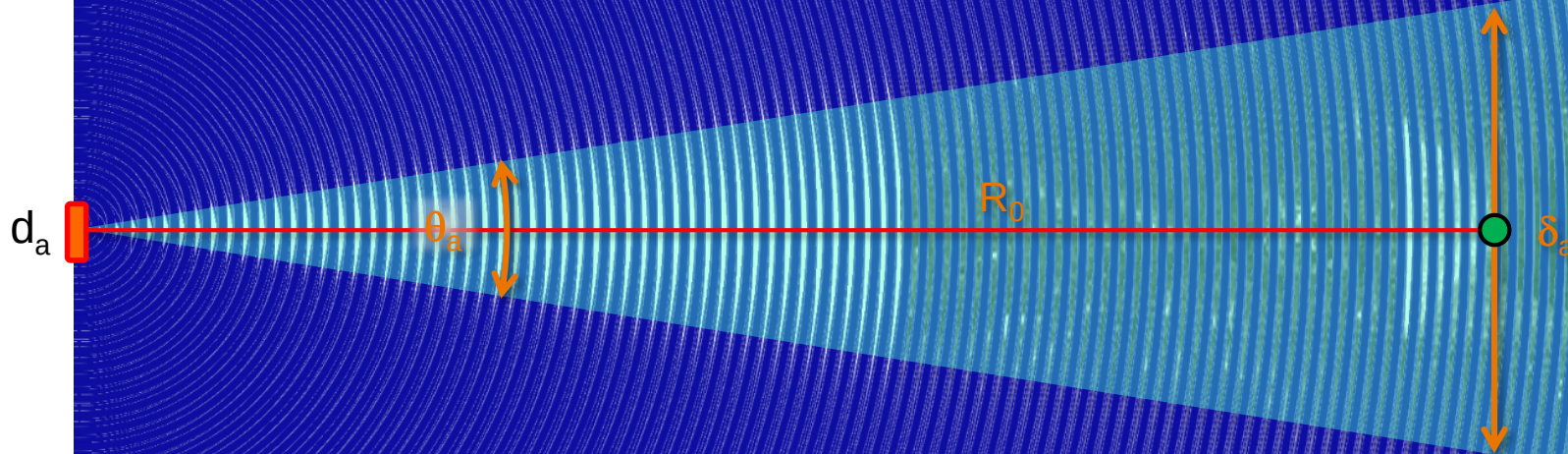
y Range



Angular resolution: $\theta_a \approx \frac{\lambda}{d_a} = \frac{\text{Wavelength}}{\text{(Real) Antenna Aperture}}$ (Diffraction)

Spatial resolution (Az): $\delta_a = \theta_a R_0 = \frac{\lambda}{d_a} R_0$ where R_0 is the distance to the scatterer

Antenna



- High azimuth resolution requires high frequencies, large antennas & small ranges.
- The azimuth resolution degrades with increasing distance R_0 to the scatterer.

This is a serious limitation for spaceborne remote sensing applications (large R_0 's)

Azimuth x
 y Range

Azimuth
x
y Range



Azimuth
x
y Range

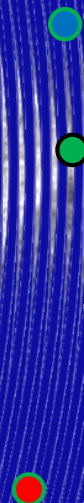


Antenna
Range

Azimuth
x
y Range



Range



Antenna
Range

Azimuth
x
y Range



Angular resolution: $\theta_a \approx \frac{\lambda}{d_a} = \frac{\text{Wavelength}}{\text{(Real) Antenna Aperture}}$

(Diffraction)

Real Antenna

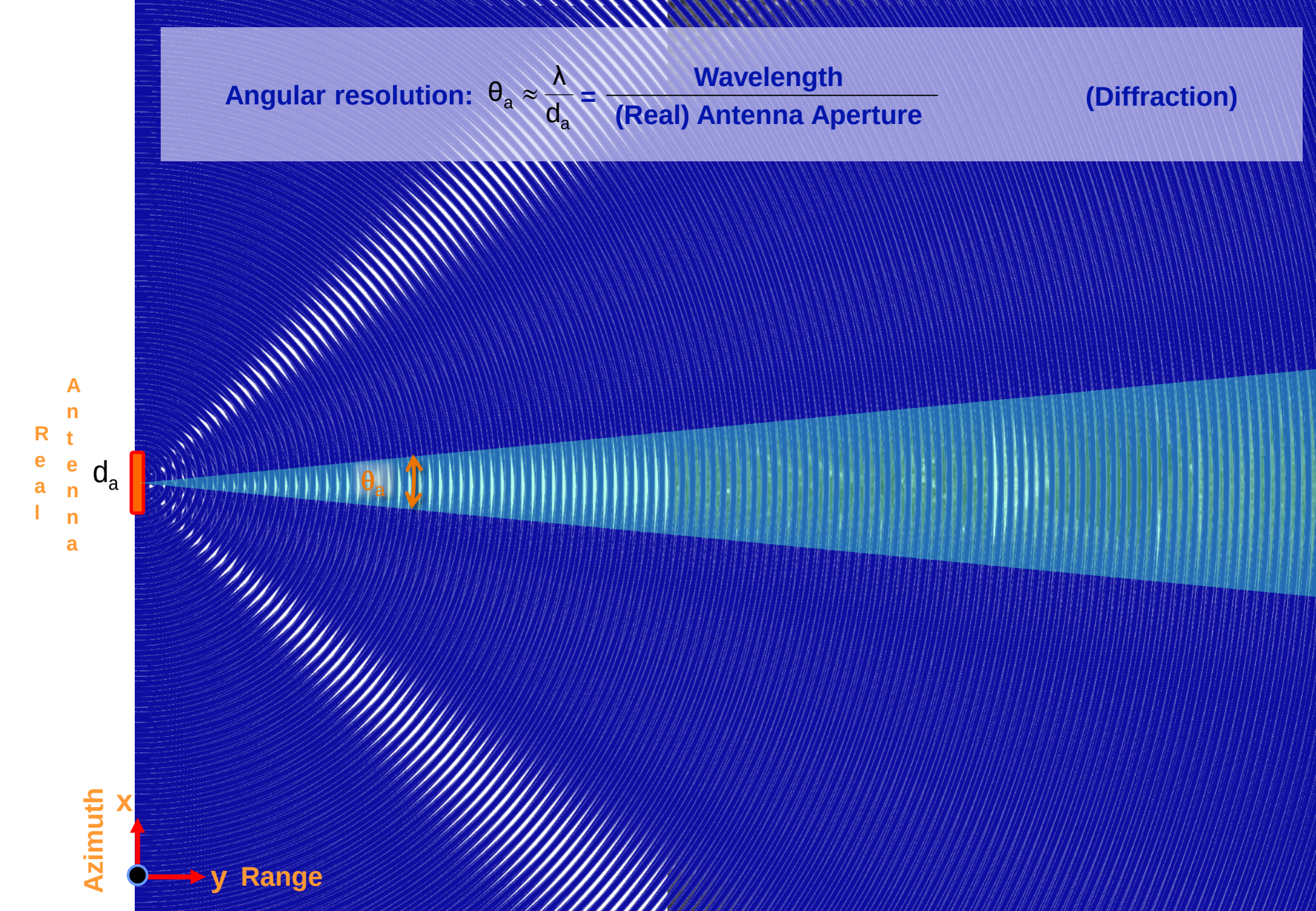
d_a

θ_a

Azimuth

x

y Range



Angular resolution: $\theta_a \approx \frac{\lambda}{d_a} = \frac{\text{Wavelength}}{\text{(Real) Antenna Aperture}}$

(Diffraction)

Antenna
Real

d_a

θ_a

Azimuth

x

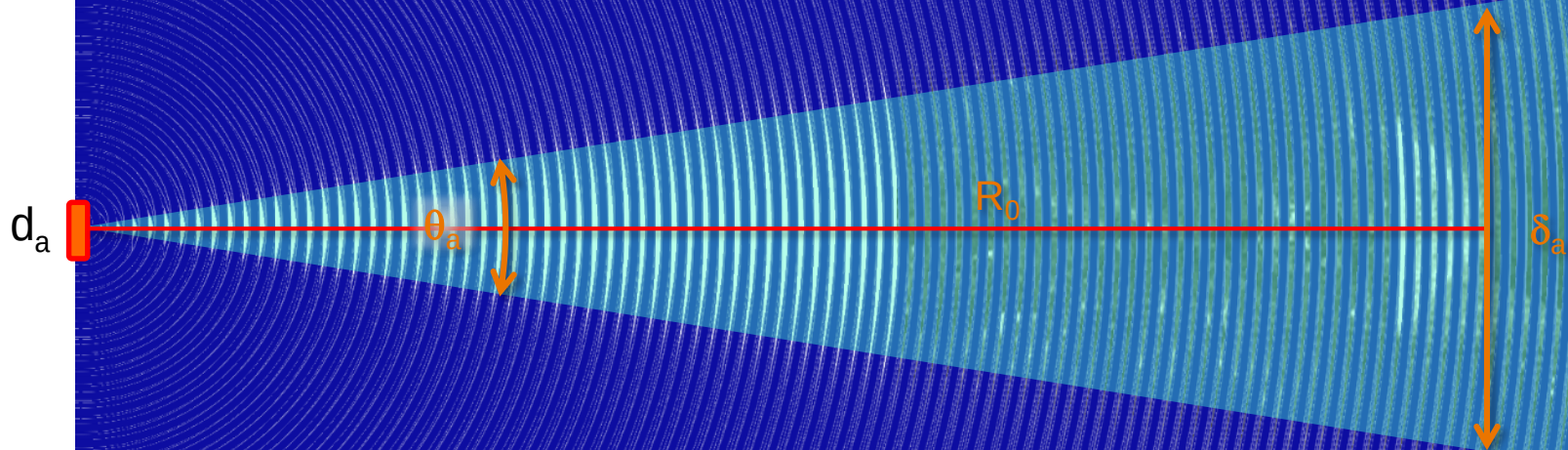
y Range



Angular resolution: $\theta_a \approx \frac{\lambda}{d_a} = \frac{\text{Wavelength}}{\text{(Real) Antenna Aperture}}$ (Diffraction)

Spatial resolution (Az): $\delta_a = \theta_a R_0 \approx \frac{\lambda}{d_a} R_0$ where R_0 is the distance to the scatterer

Antenna



► High azimuth resolution requires high frequencies, large antennas & small ranges.

► The azimuth resolution degrades with increasing distance R_0 to the scatterer.

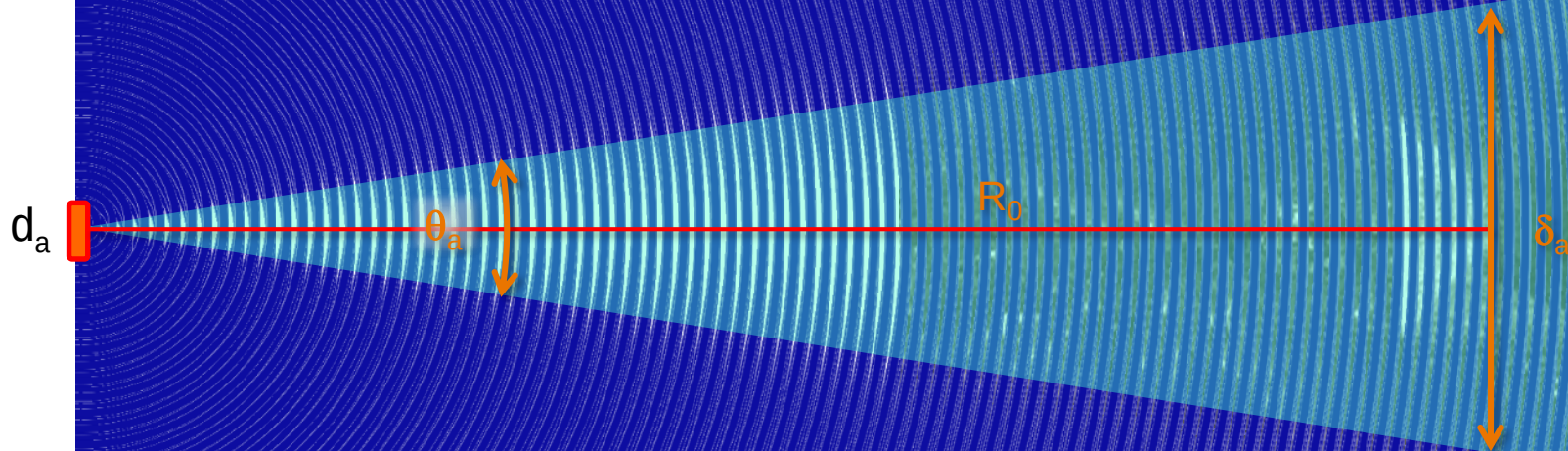
This is a serious limitation for spaceborne remote sensing applications (large R_0 's)

Azimuth x
 y Range

Angular resolution: $\theta_a \approx \frac{\lambda}{d_a} = \frac{\text{Wavelength}}{\text{(Real) Antenna Aperture}}$ (Diffraction)

Spatial resolution (Az): $\delta_a = \theta_a R_0 \approx \frac{\lambda}{d_a} R_0$ where R_0 is the distance to the scatterer

Antenna



Example 1: X-Band Airborne system, 50MHz bandwidth ($\tau=20\text{ns}$), 3m antenna, 3000m range

$$\delta_r = 3 \text{ m}$$

$$\delta_a = 30 \text{ m}$$

Example 2: X-Band Satellite system, 50MHz bandwidth ($\tau=20\text{ns}$), 12m antenna, 800km range

$$\delta_r = 3 \text{ m}$$

$$\delta_a = 2000 \text{ m !}$$

Azimuth

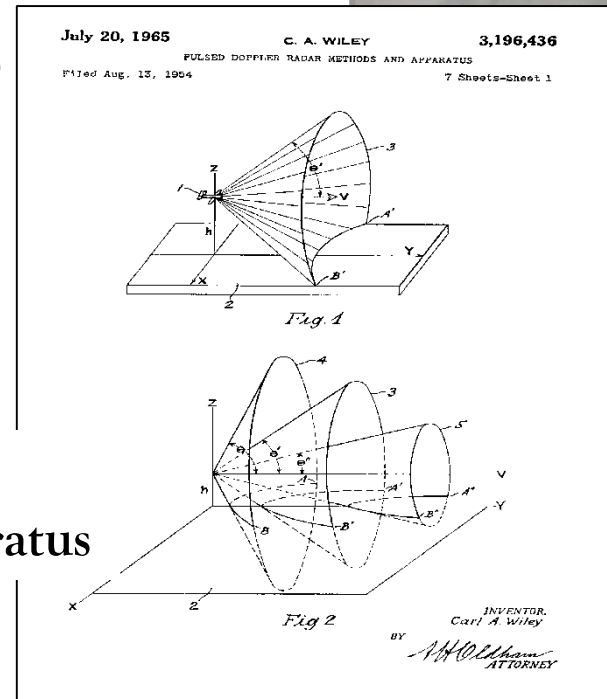
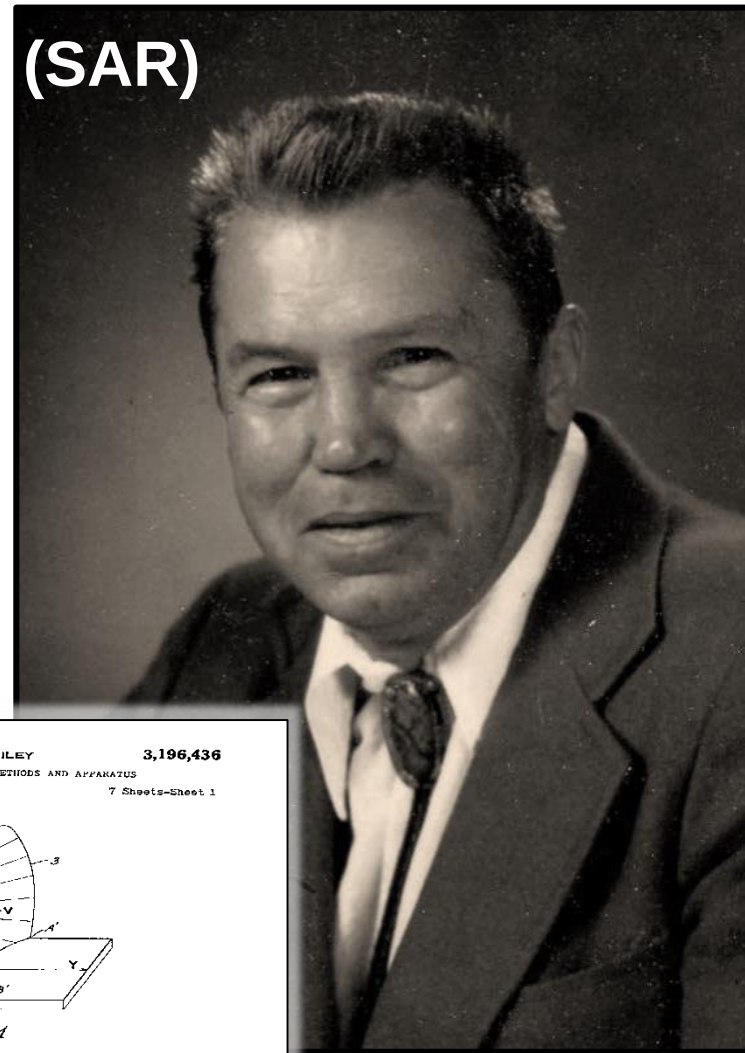
x

y Range

The Solution: Synthetic Aperture Radar (SAR)

Carl A. Wiley, 1951:

- **A large antenna** - required for achieving a high spatial resolution in azimuth - is “synthesised” using an array of small antennas working together.
- This array of small antennas is formed by using **a real aperture antenna**
 - mounted on a moving (with velocity v_p) **platform**
 - operated in a **pulsed mode** (with a Pulse Repetition Frequency given by PRF).



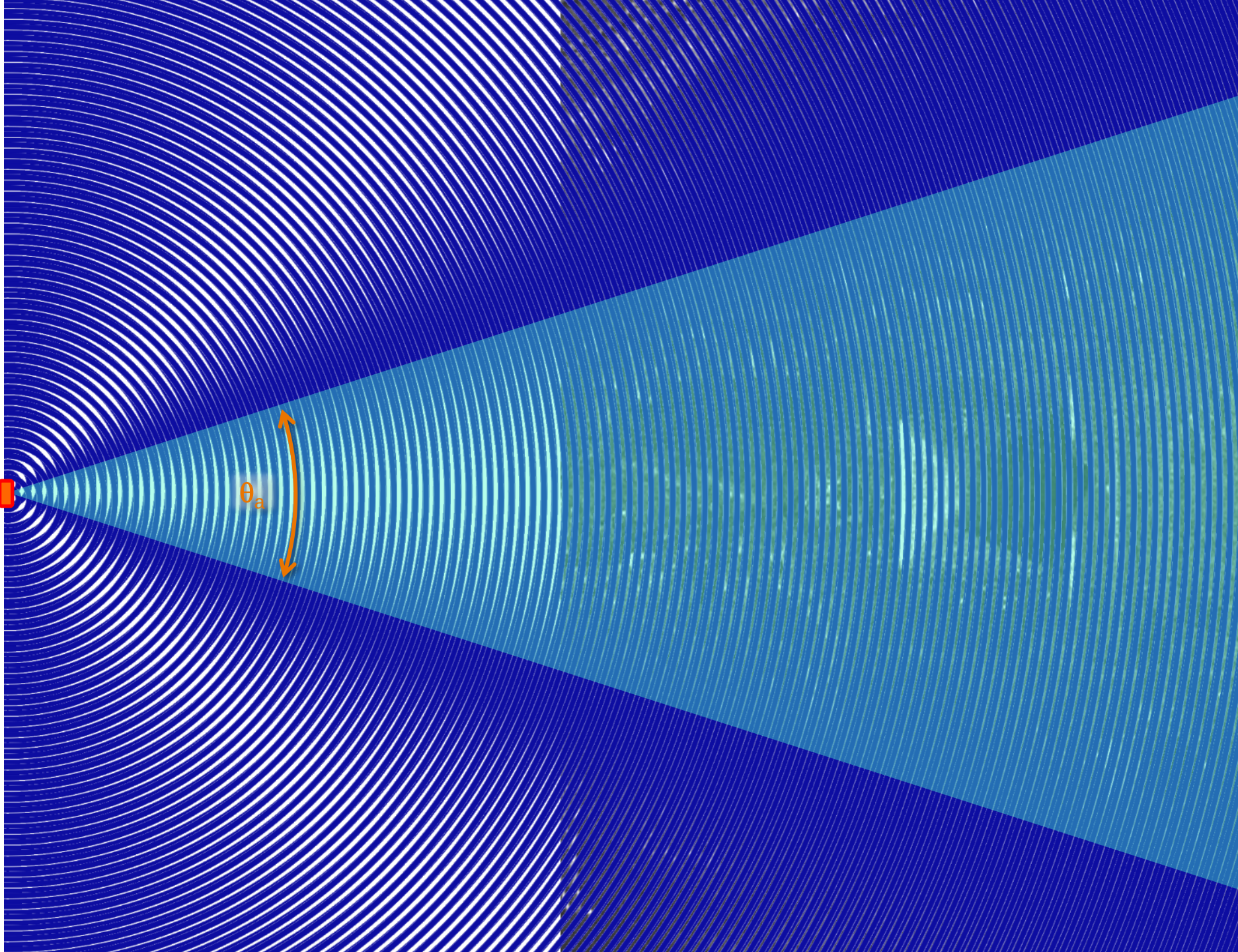
C.A. Wiley / US Patent 3,196,436 / 1951
Pulsed Doppler Radar Methods and Apparatus

A
p
e
r
t
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r
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R
e
a
l

d_a

θ_a



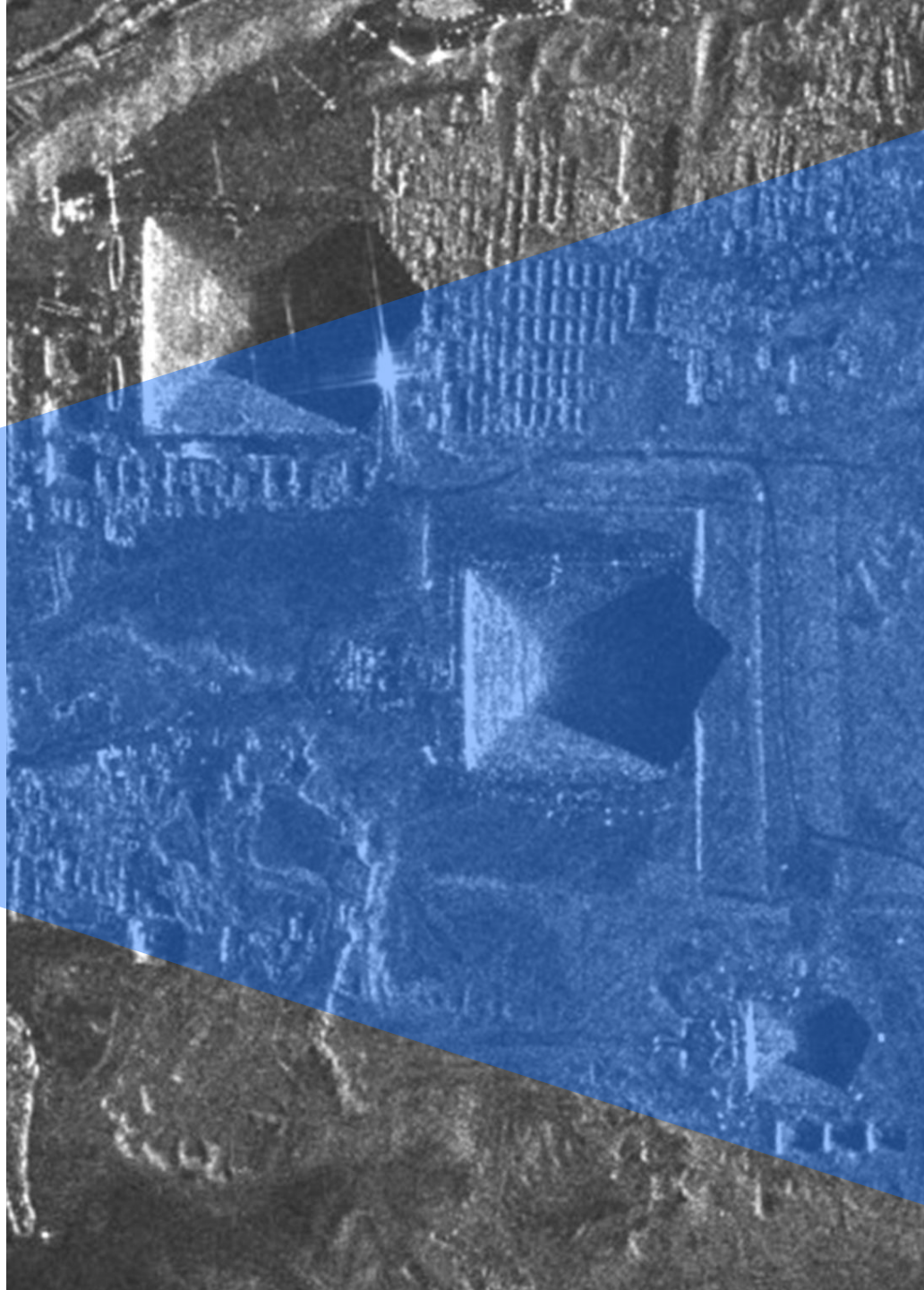
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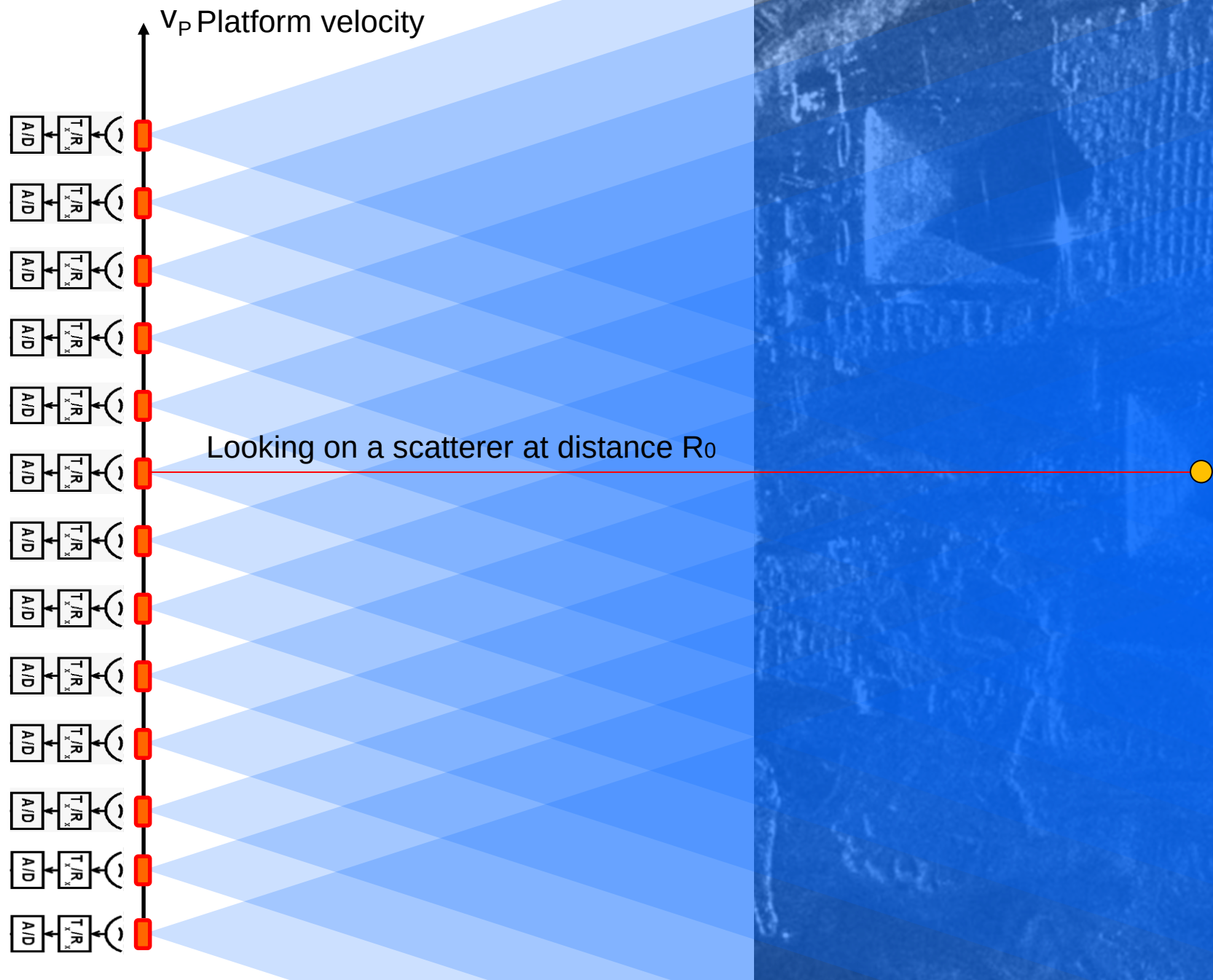
R
e
a
l

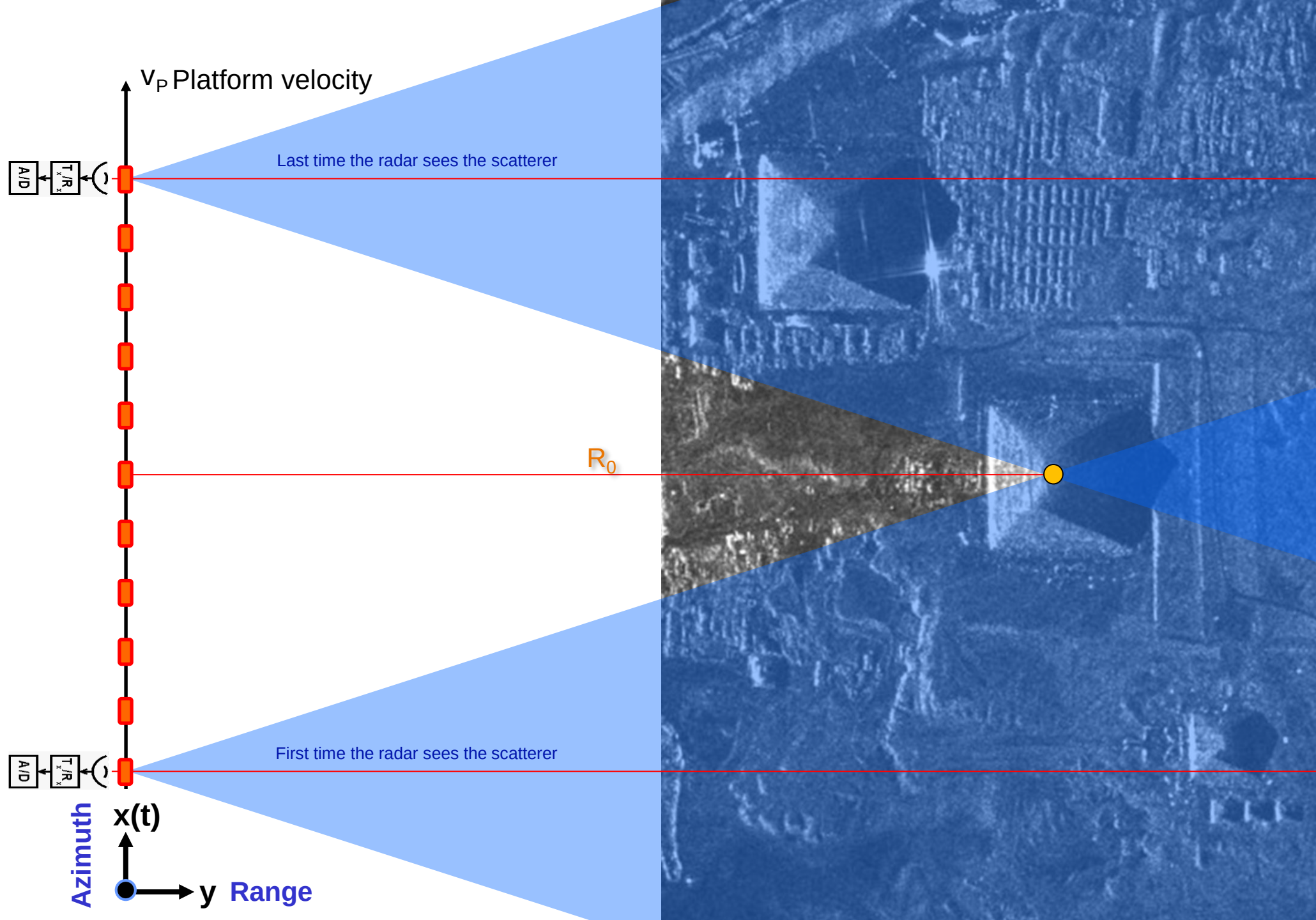
d_a

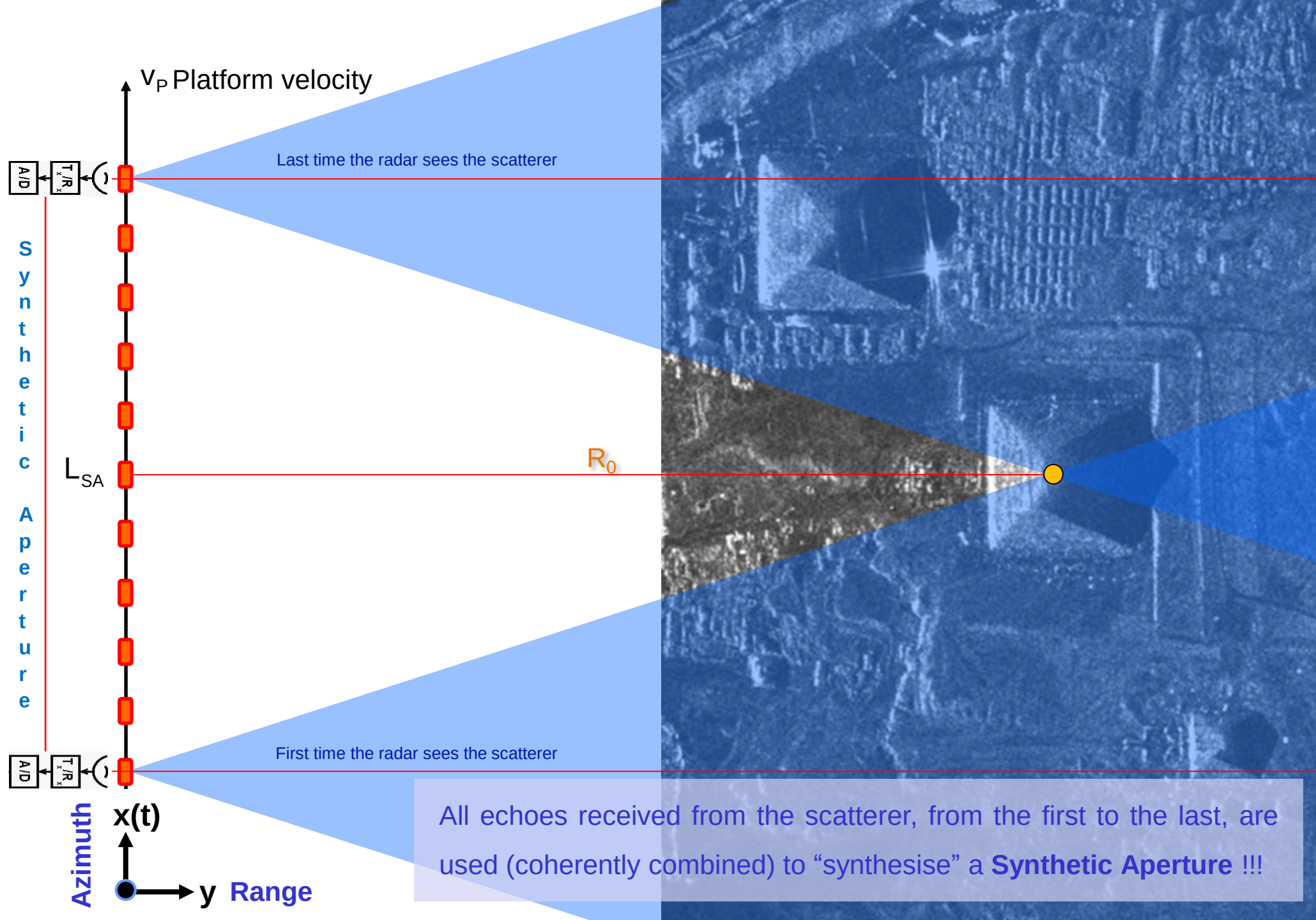


θ_a

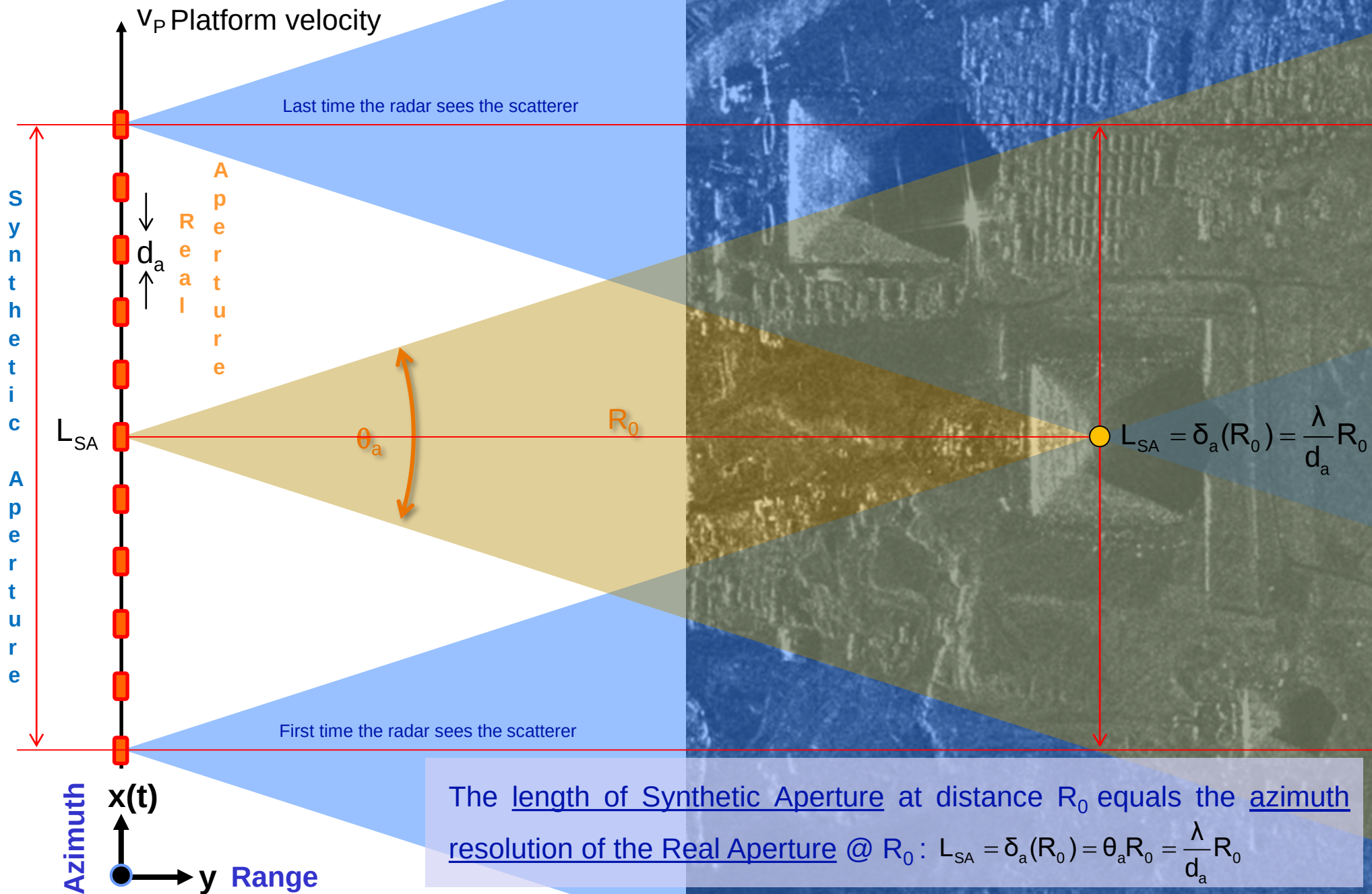


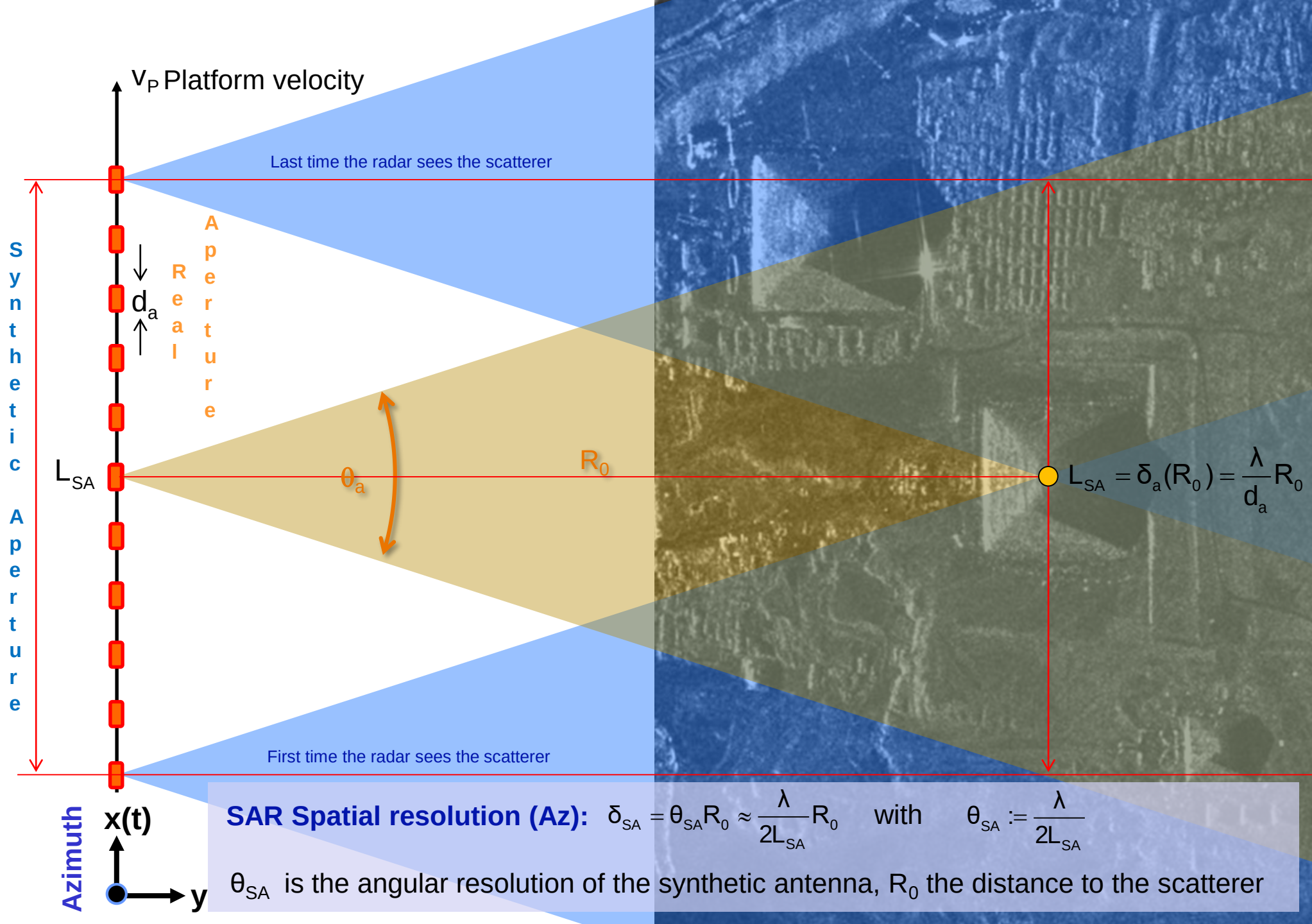


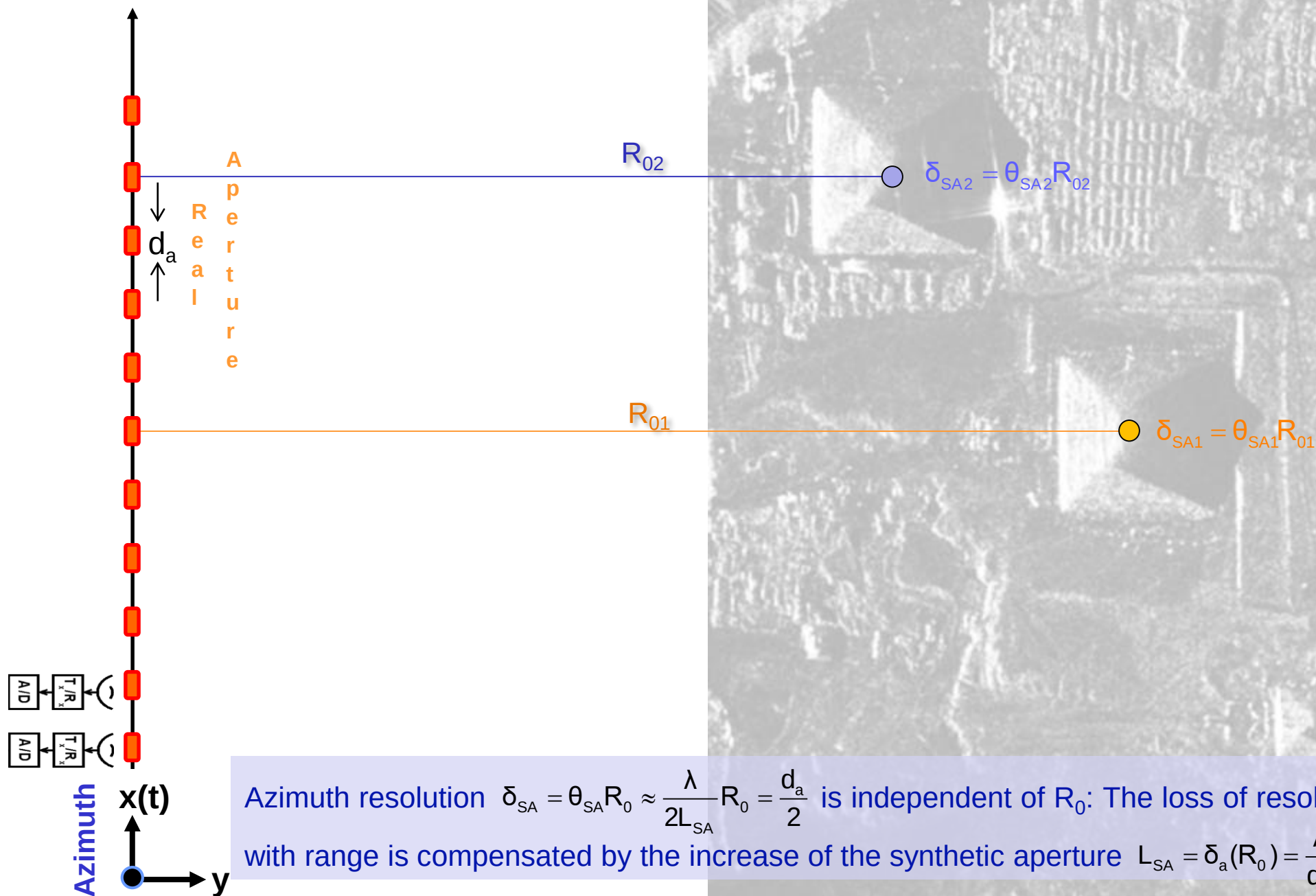


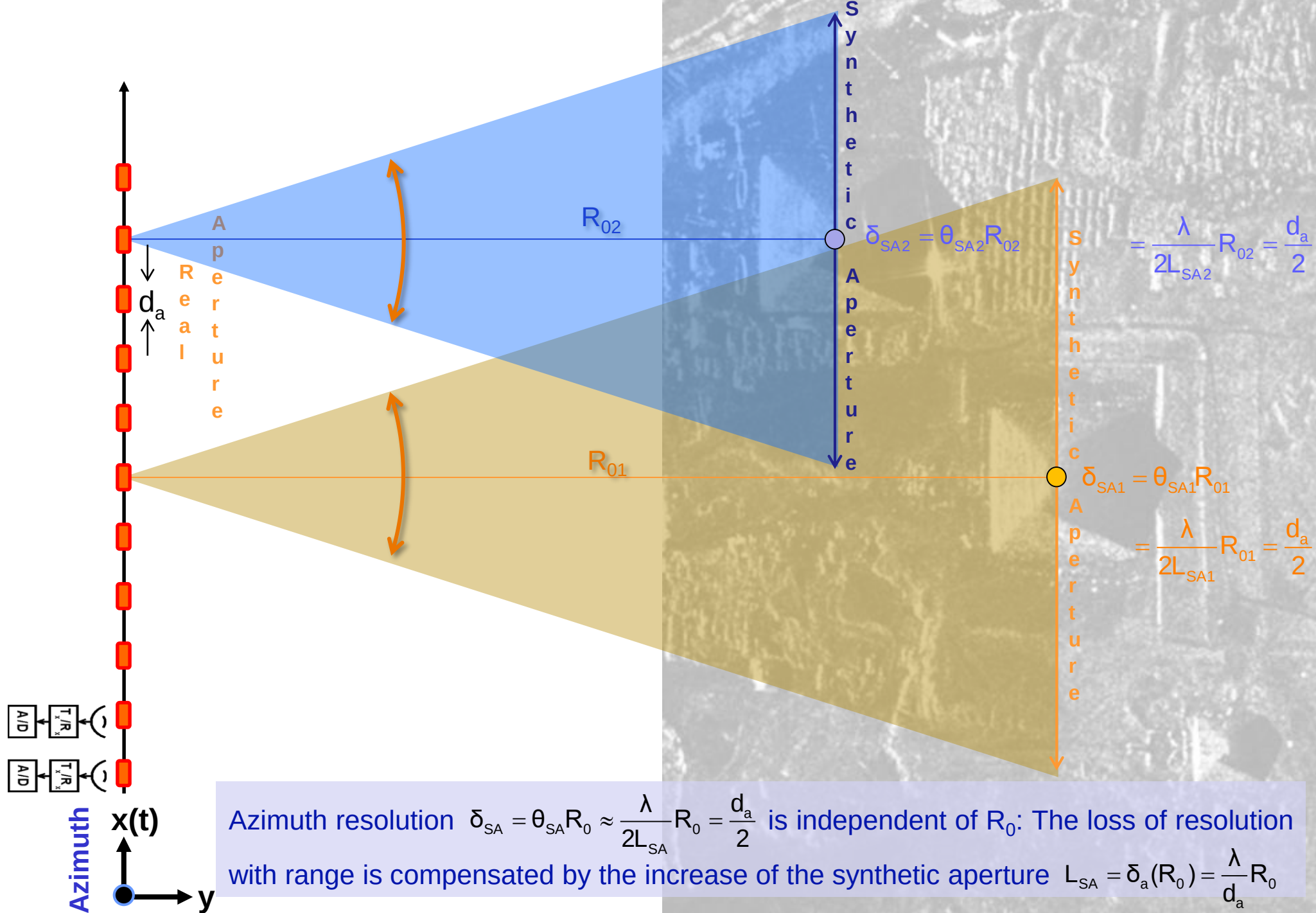


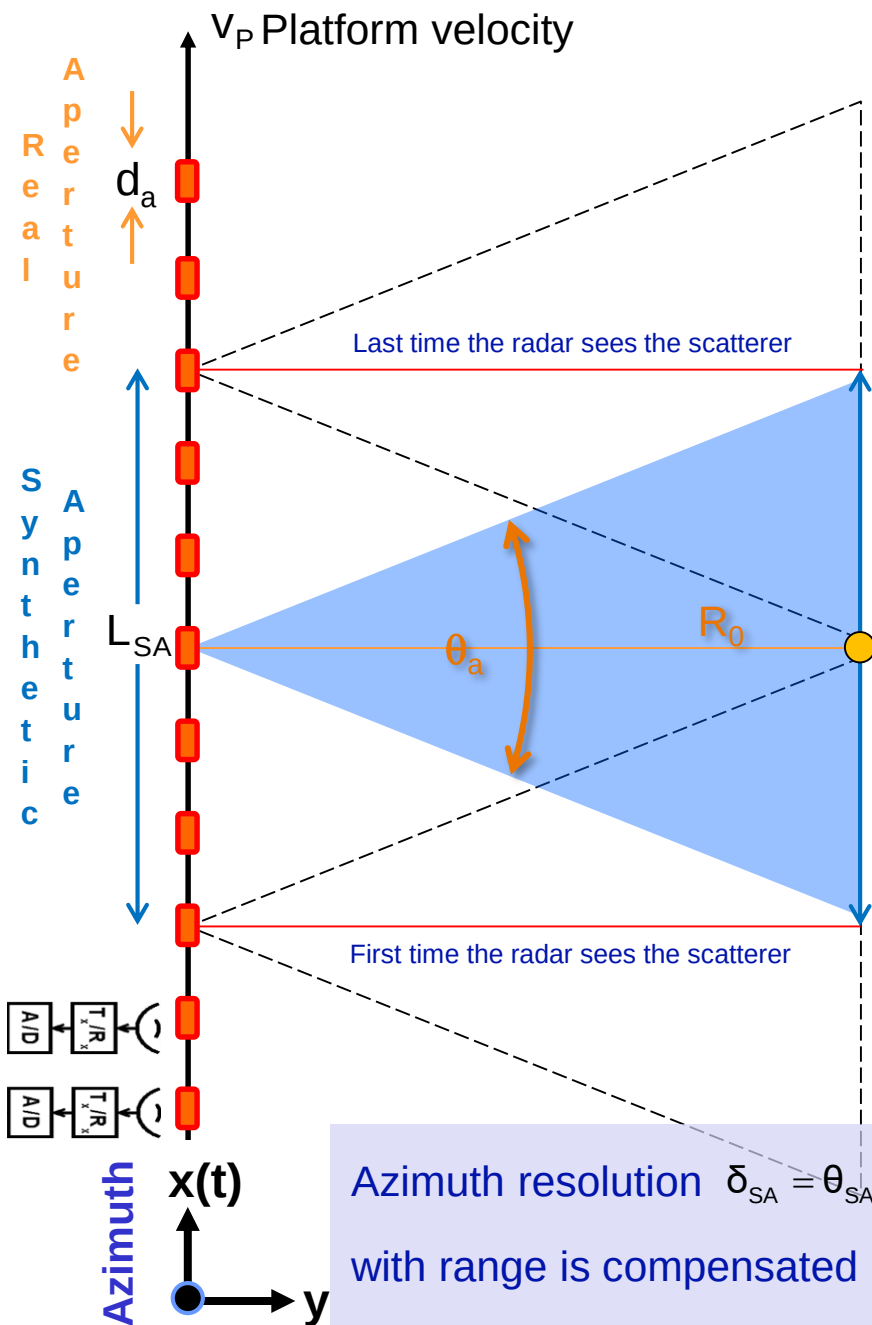
All echoes received from the scatterer, from the first to the last, are used (coherently combined) to “synthesise” a **Synthetic Aperture** !!!











- Angular resolution of synthetic antenna:

$$\theta_{SA} = \frac{\lambda}{2L_{SA}} = \frac{d_a}{2R_0}$$

decreases with distance R_0

- Length of the synthetic aperture (at distance R_0):

$$L_{SA} = \delta_a = \theta_a R_0 = \frac{\lambda}{d_a} R_0$$

increases with R_0

- Illumination time of an scatterer at distance R_0 :

$$t_{SA} = \frac{L_{SA}}{v_P} = \frac{\lambda}{d_a} \frac{R_0}{v_P}$$

increases with R_0

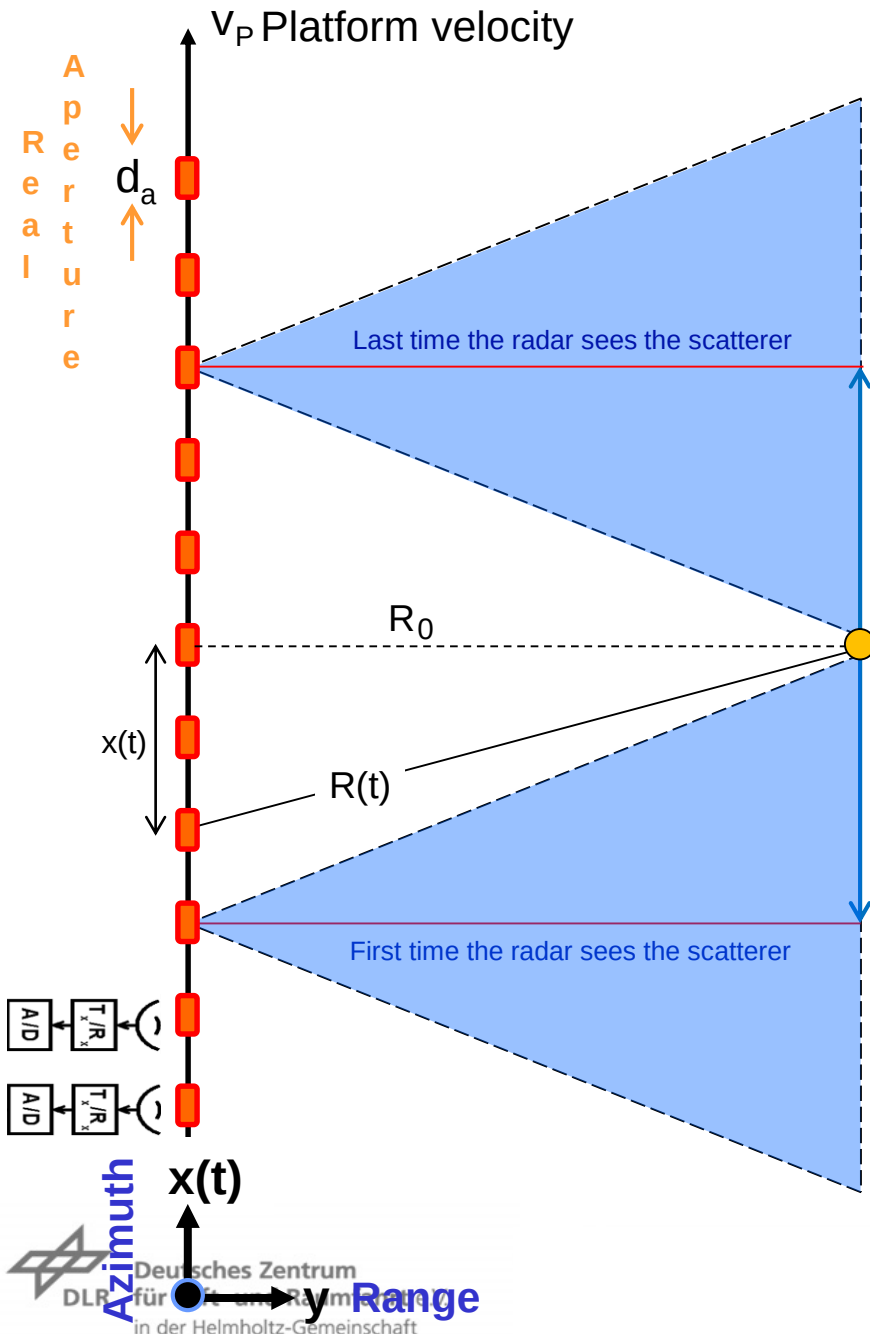
- Azimuth resolution: δ_a

$$\delta_{SA} = \theta_{SA} R_0 = \frac{d_a}{2}$$

v_P : velocity of the platform

= Half (Real) Antenna Length!

Azimuth resolution $\delta_{SA} = \theta_{SA} R_0 \approx \frac{\lambda}{2L_{SA}} R_0 = \frac{d_a}{2}$ is independent of R_0 : The loss of resolution with range is compensated by the increase of the synthetic aperture $L_{SA} = \delta_a(R_0) = \frac{\lambda}{d_a} R_0$



- Radar-Scatterer distance variation along the aperture:

$$R(t) = \sqrt{R_0^2 + x(t)^2}$$

$$= R_0 \sqrt{1 + \frac{x(t)^2}{R_0^2}} \approx R_0 \left(1 + \frac{x(t)^2}{2R_0^2} + \dots \right) \approx R_0 + \frac{x(t)^2}{2R_0}$$

$$R(t) = R_0 + \frac{v_P^2 t^2}{2R_0} \quad \text{with} \quad x(t) = v_P t$$

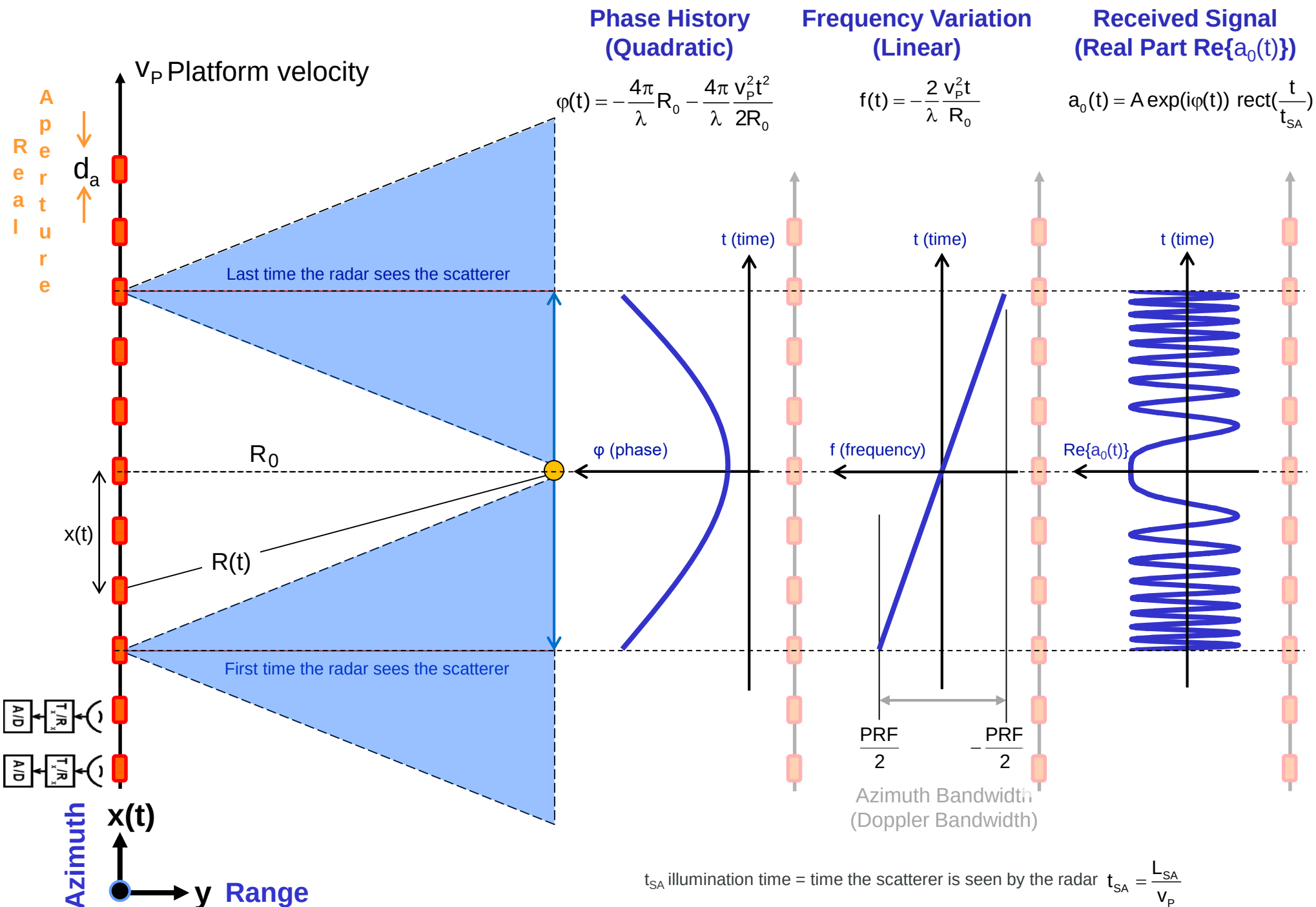
- Phase variation (e.g. "Phase history") along the aperture:

$$\varphi(t) = -2 \frac{2\pi}{\lambda} R(t) = -\frac{4\pi}{\lambda} R_0 - \frac{4\pi}{\lambda} \frac{v_P^2 t^2}{2R_0}$$

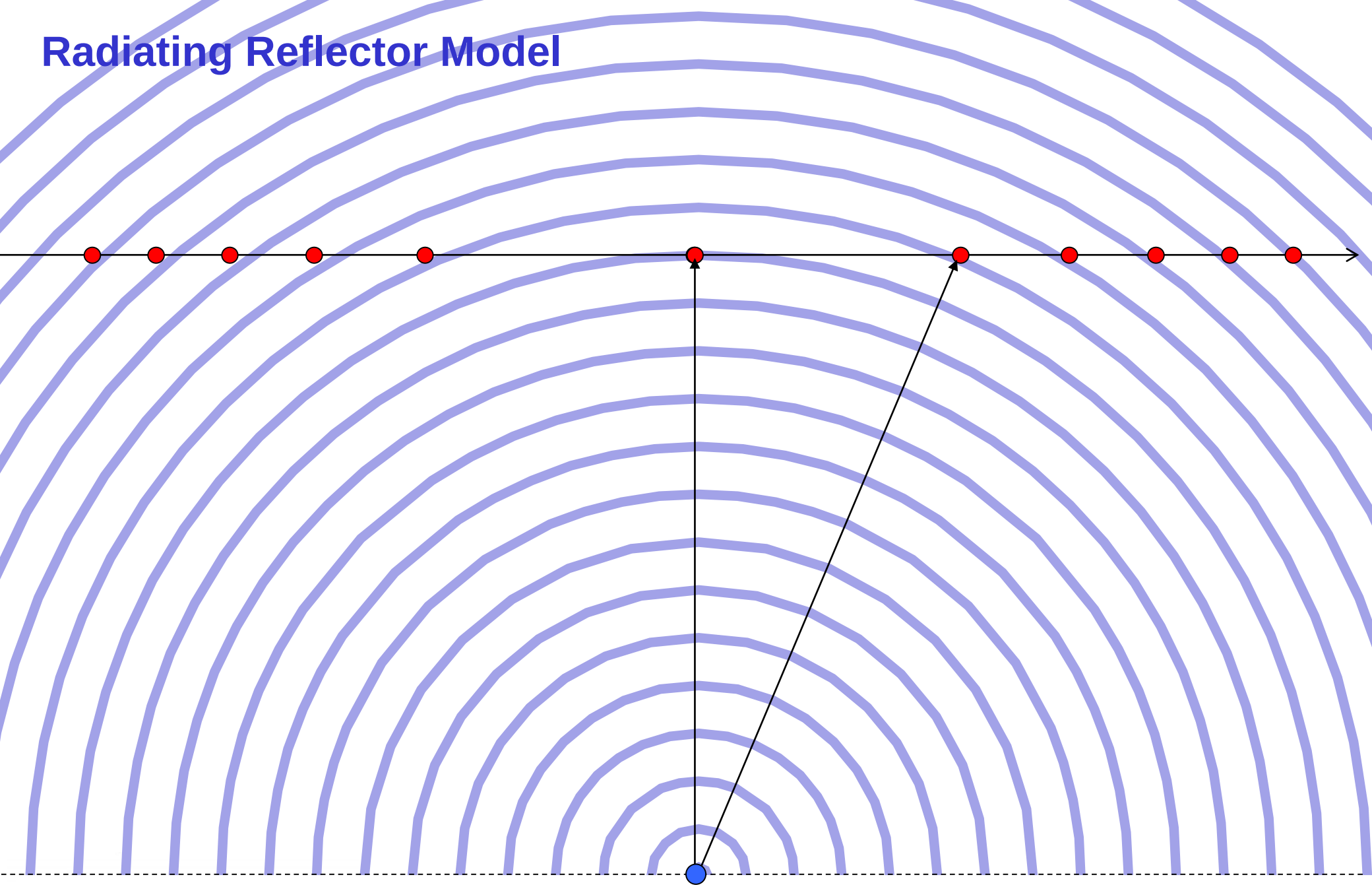
- Frequency variation along the aperture:

$$f(t) = \frac{1}{2\pi} \frac{d\varphi}{dt} = -\frac{2}{\lambda} \frac{v_P^2 t}{R_0}$$

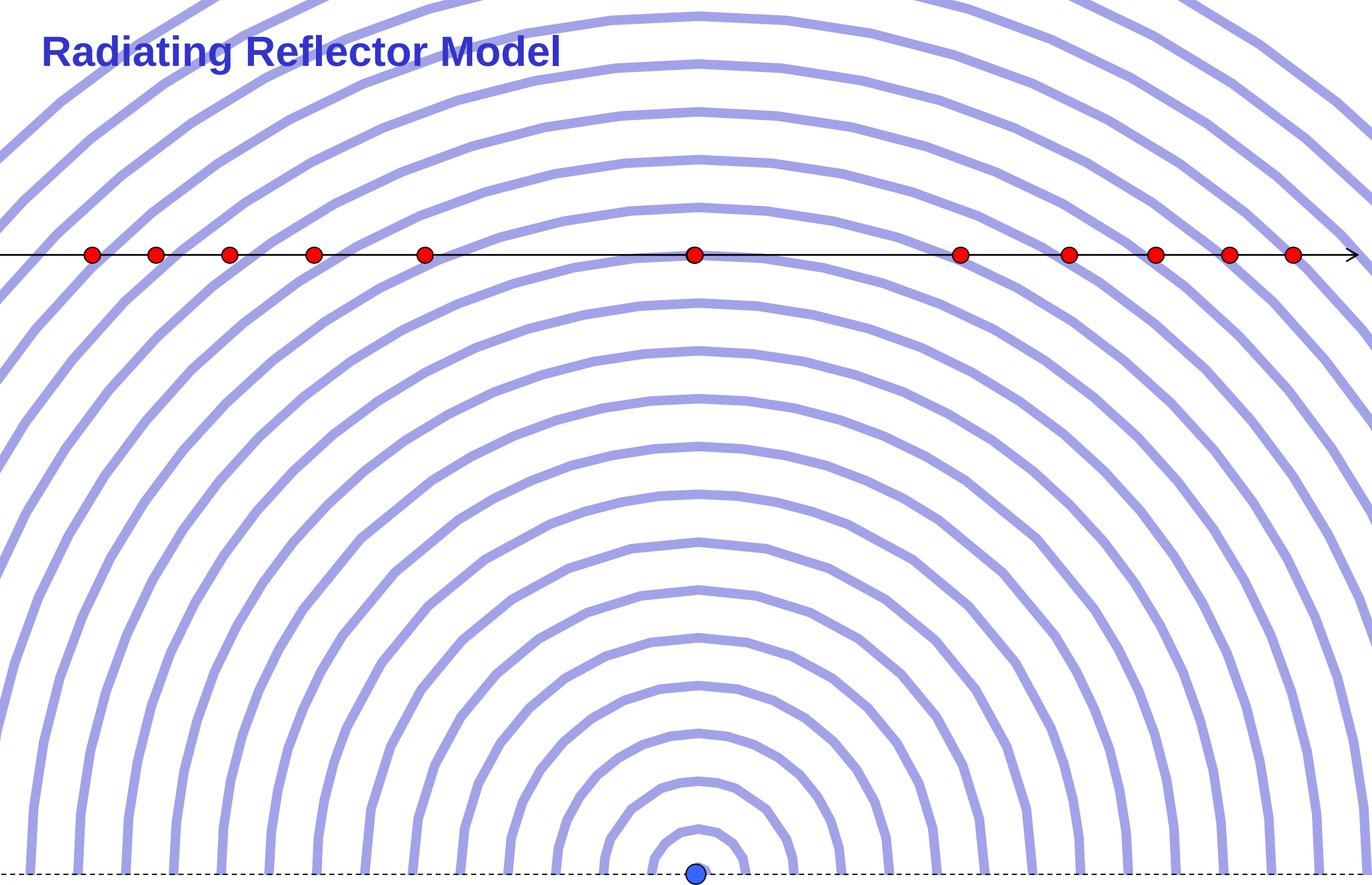
Range (R_0)
Dependency



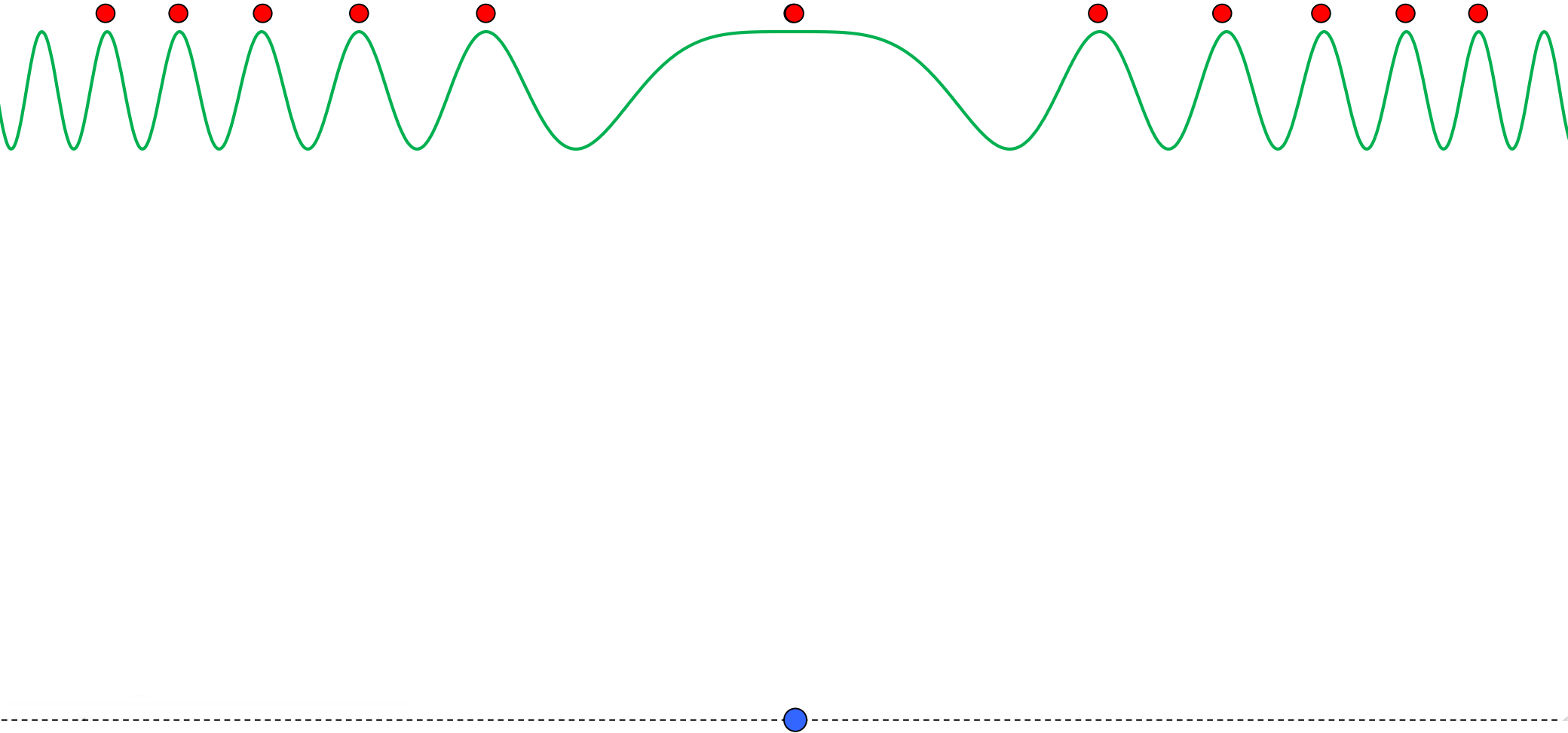
Radiating Reflector Model

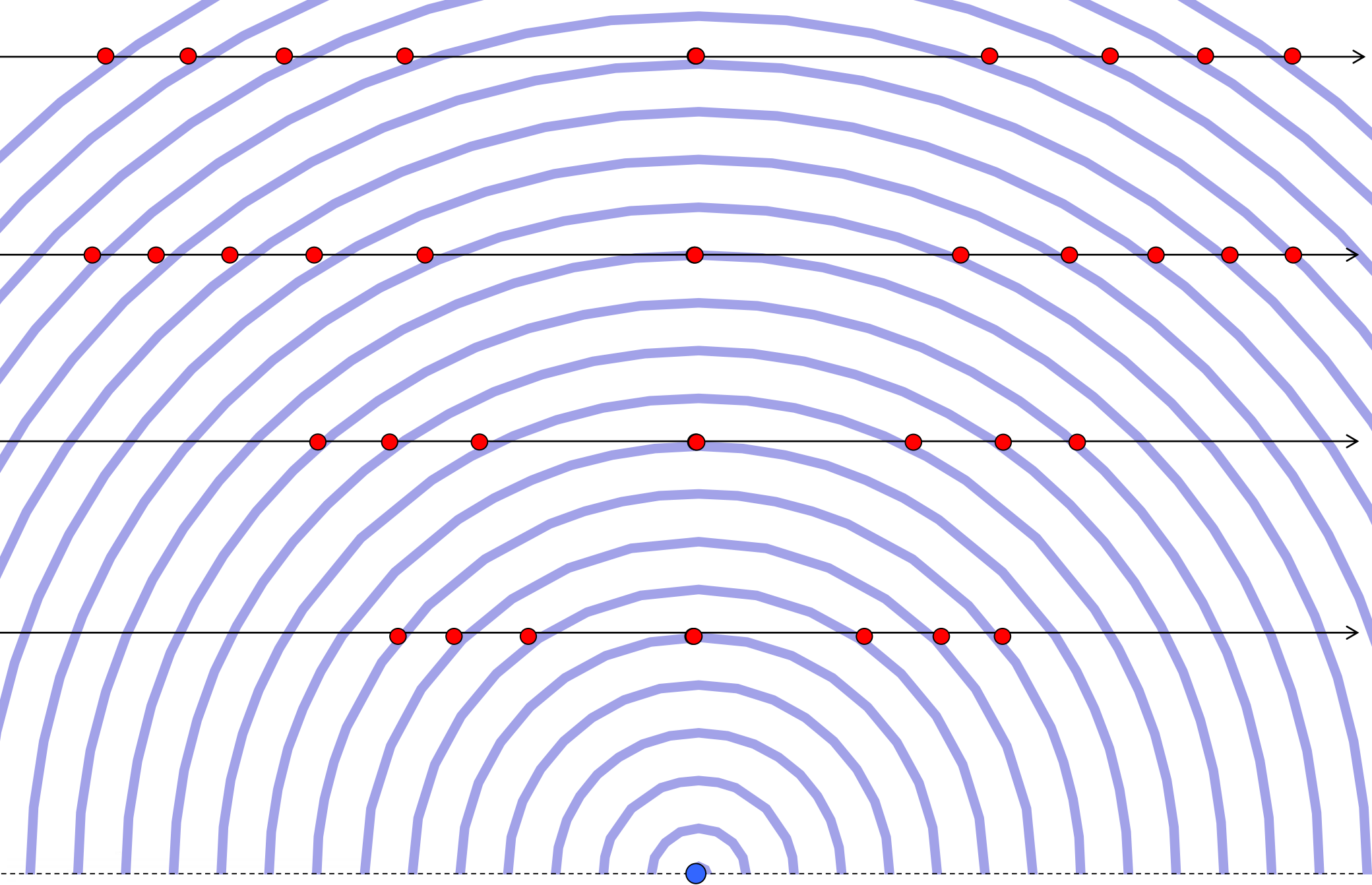


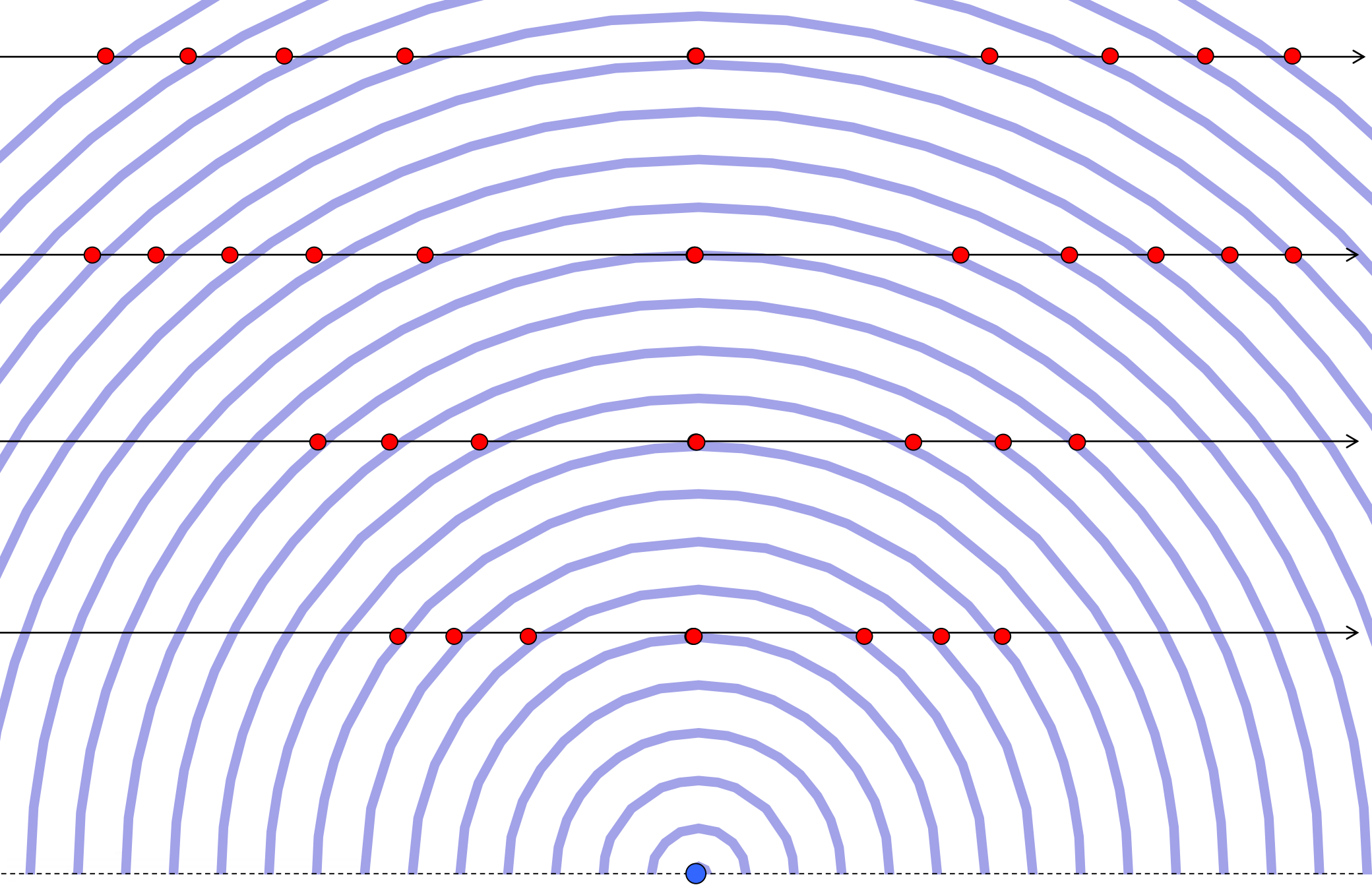
Radiating Reflector Model

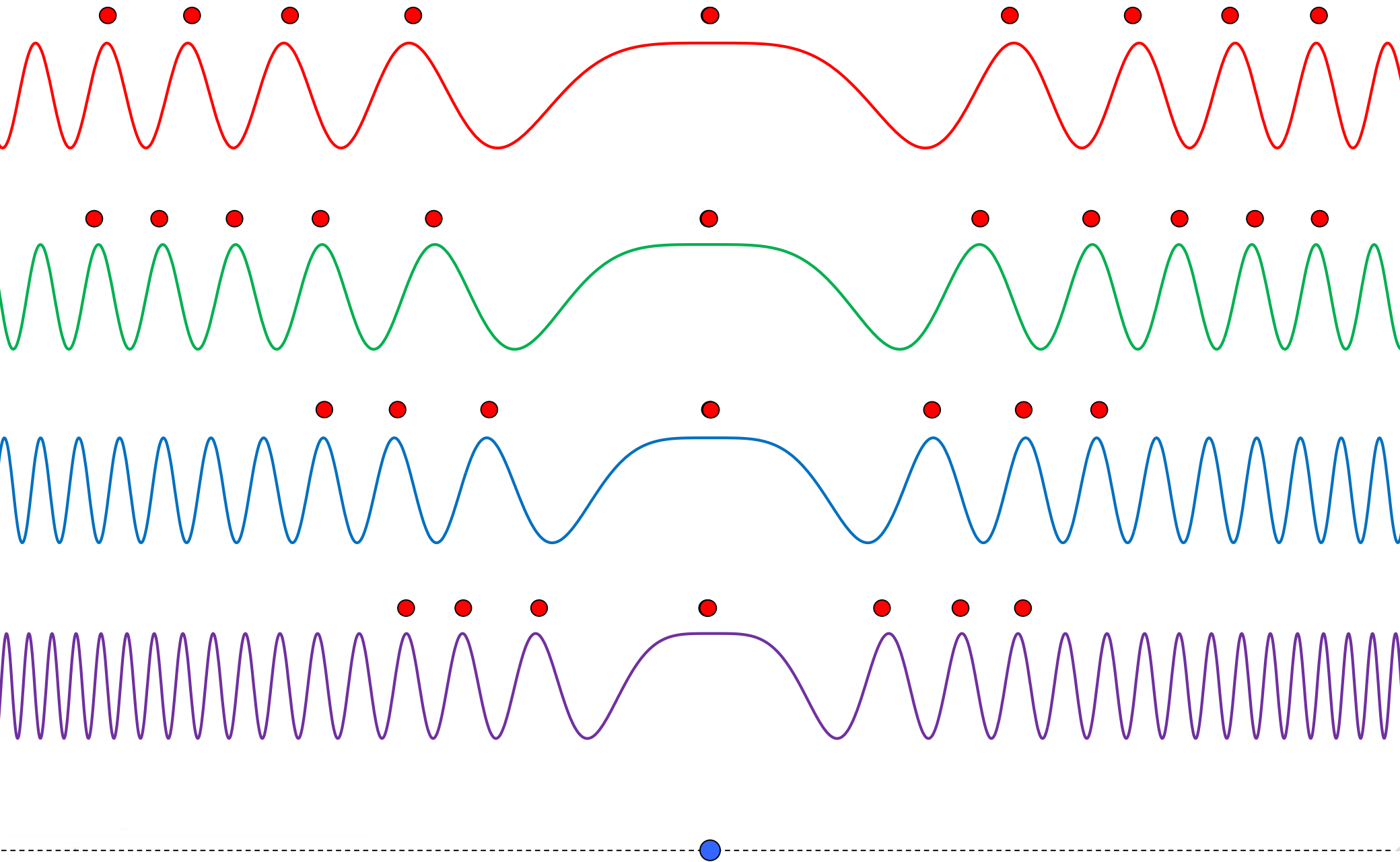


Radiating Reflector Model



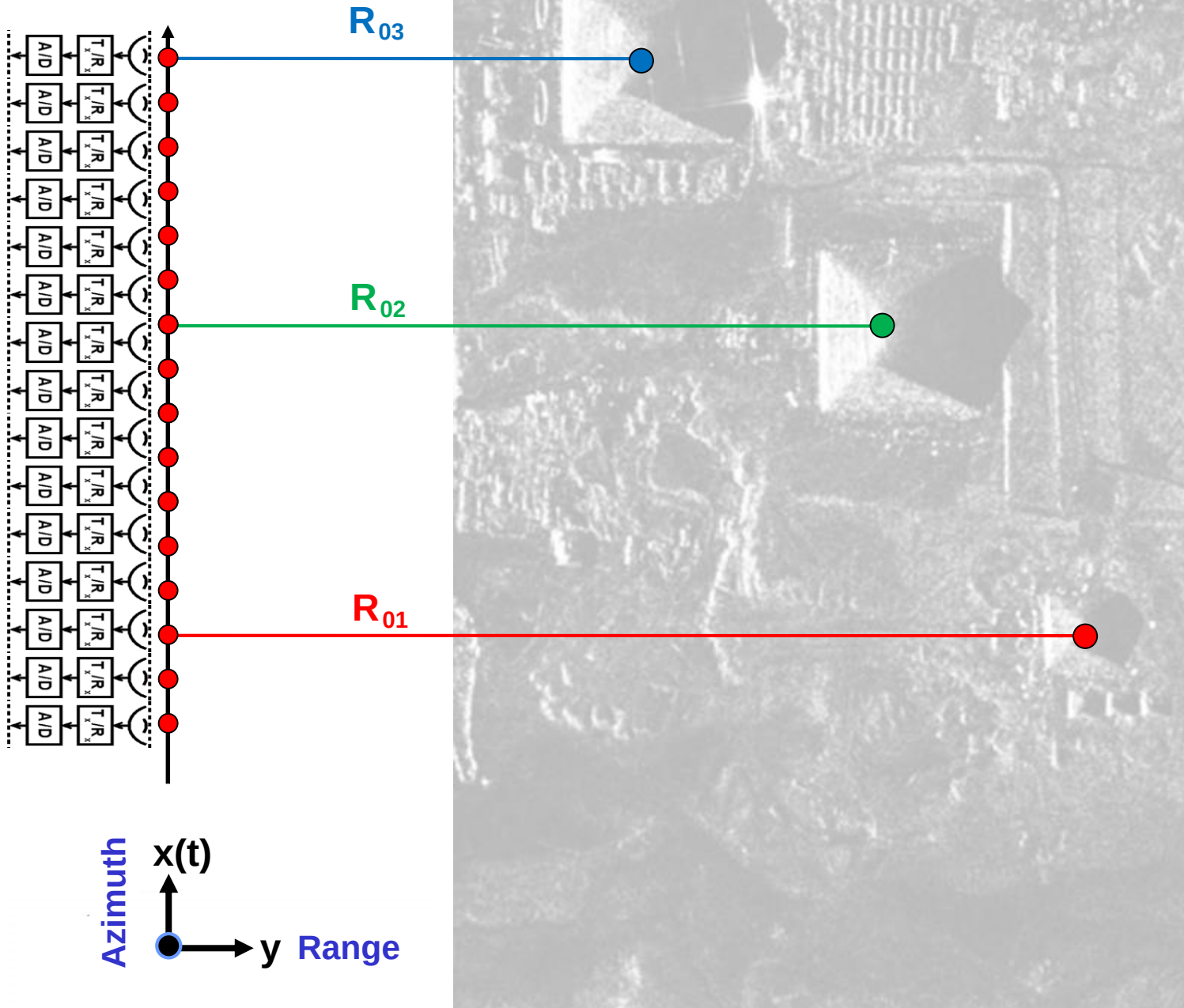




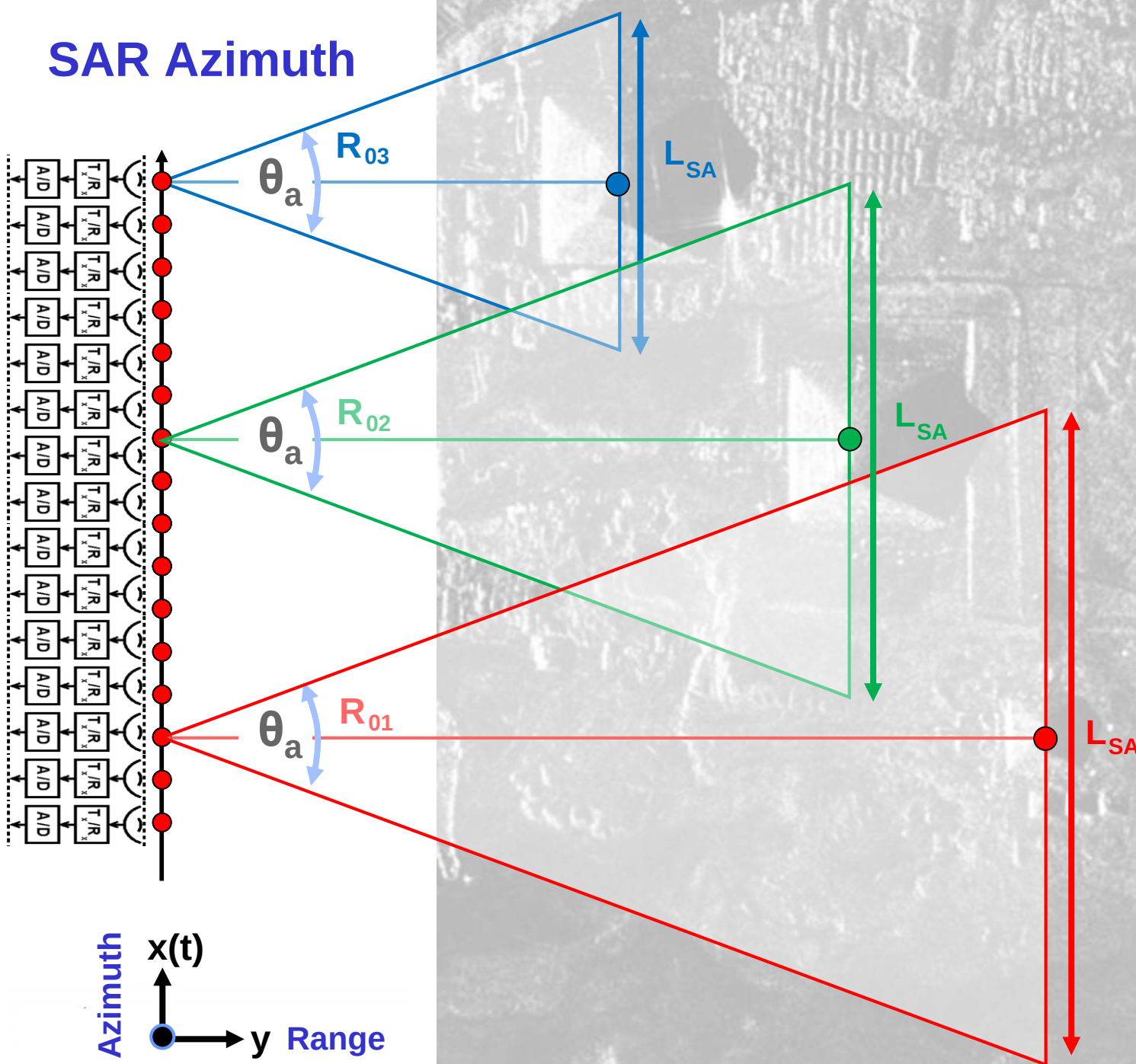


SAR Azimuth

Resolution



SAR Azimuth



Resolution

The deterioration of spatial resolution δ_{SA} with increasing range distance R_0

$$\delta_{SA} = \theta_{SA} R_0 = \frac{\lambda}{2 L_{SA}} R_0$$

is compensated with the increase of the synthetic aperture L_{SA}

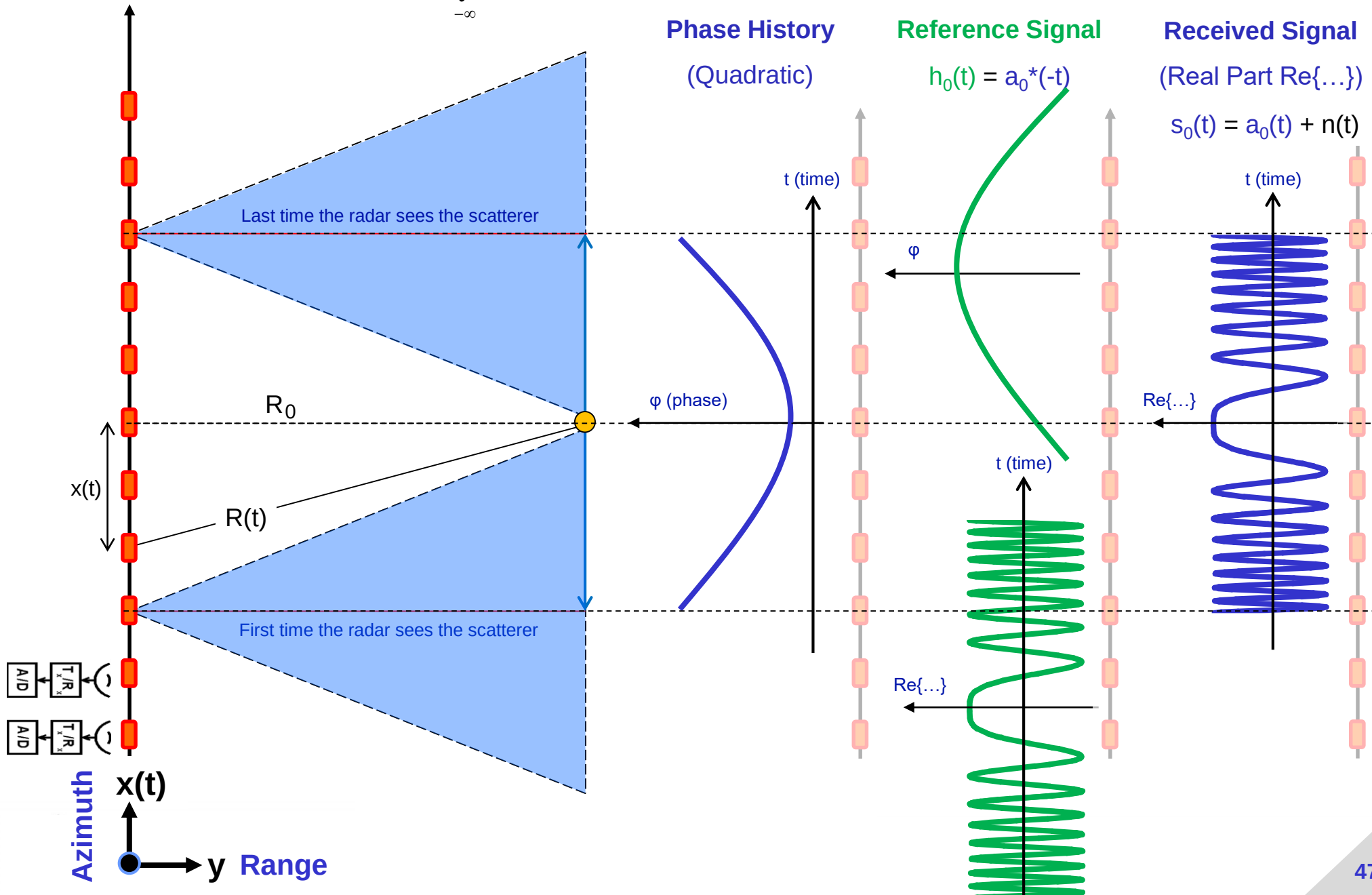
$$L_{SA} = \frac{\lambda}{d_a} R_0$$

equivalent to the increase of the time t_{SA} the scatterer is seen by the SAR:

$$t_{SA} = \frac{L_{SA}}{v_P} = \frac{\lambda}{d_a v_P} R_0$$

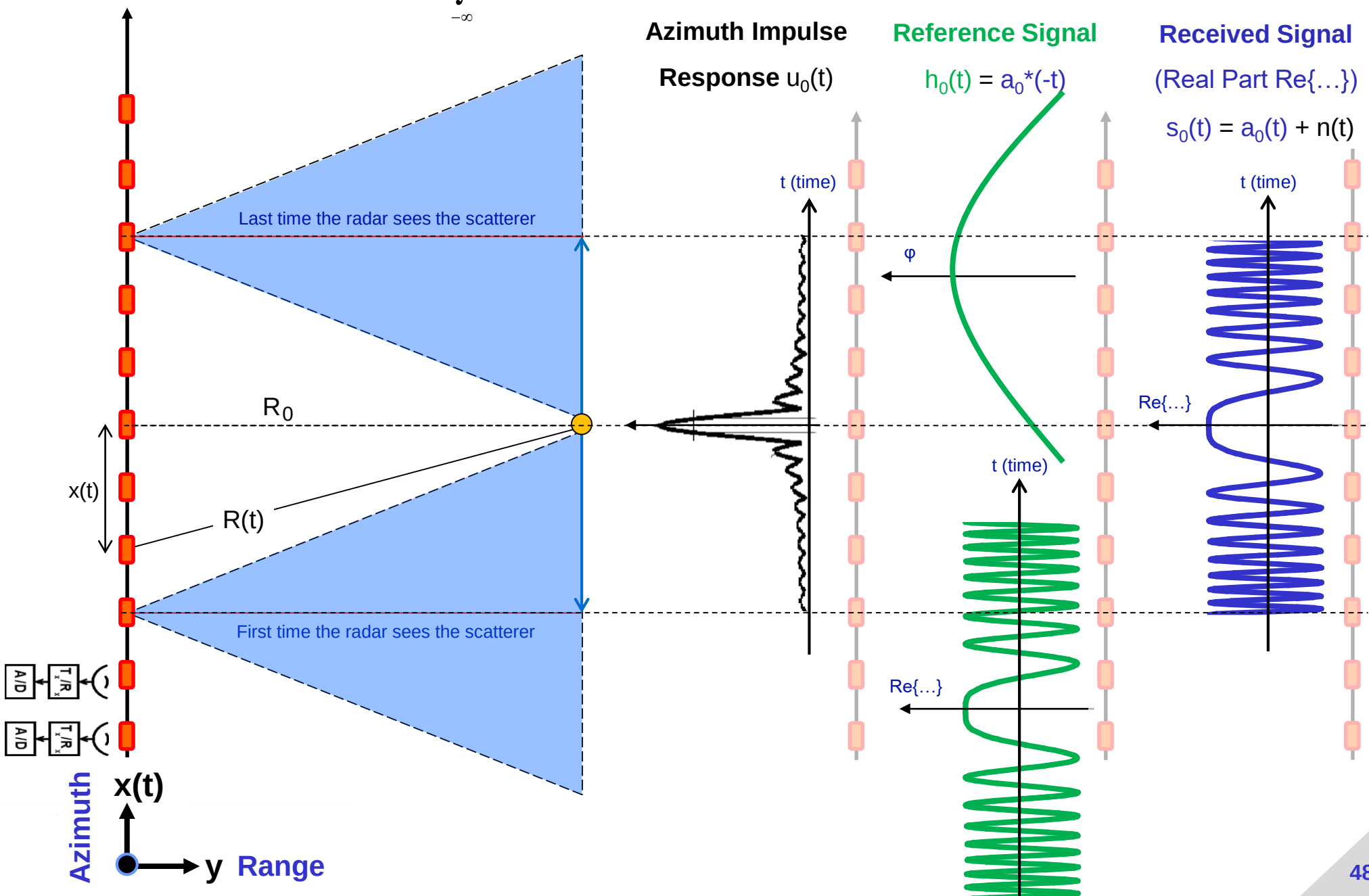
Matched Filter: $u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot h_0(t - T) dT$

SAR Azimuth Compression



Matched Filter: $u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot h_0(t - T) dT$

SAR Azimuth Compression



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Part 2: Azimuth Resolution

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